English

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Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
AIOT
12-05-2024

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $\mathbf{g = 10\ m/s^2}$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS		
Atomic No.	:	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses	:	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

•	Boltzmann constant	$k = 1.38 \times 10^{-23}\ JK^{-1}$
•	Coulomb's law constant	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
•	Universal gravitational constant	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
•	Speed of light in vacuum	$c = 3 \times 10^{8}\ ms^{-1}$
•	Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
•	Wien's displacement law constant	$b = 2.89 \times 10^{-3}\ m\text{–}K$
•	Permeability of vacuum	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
•	Permittivity of vacuum	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
•	Planck constant	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

This section contains **SIX (06)** questions.
Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

For Example : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. A variable capacitor with initial capacitance C_0 is charged to a voltage V_0 and is short circuited through a resistance R. If current remains constant during discharging (an external agent is pulling the plates apart) then choose correct option(s).

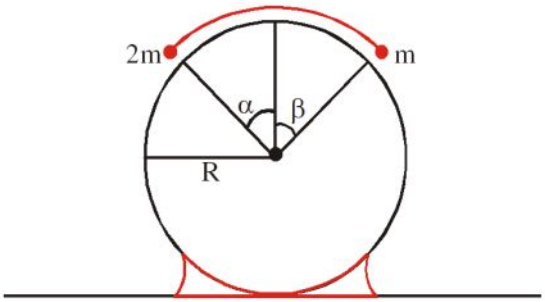
(A) Variation of capacitance with time is given by $C = C_0\left(1 - \frac{t}{RC_0}\right)$.

(B) Time taken to discharge the capacitor is given by $t = RC_0$.

(C) Total heat produced in the resistor is $C_0V_0^2$.

(D) Initial energy stored in the capacitor is $\frac{1}{2}C_0V_0^2$.

2. A cylindrical body of radius R is fixed in a horizontal position. A piece of ideal thread of length ℓ was laid on its slippery surface. On one end of thread a point like body of mass m is attached, while to the other end another point like body of mass 2m is attached . Choose correct statement(s). The system is in equilibrium.



(A) $2\sin\alpha = \sin\beta$

(B) $2\sin\beta = \sin\alpha$

(C) Maximum possible length of string is $\ell_{max} = 2\pi R/3$.

(D) Maximum possible length of string is $\ell_{max} = 2R$.

3. By properly combining two prisms made of different materials, it is possible to :-

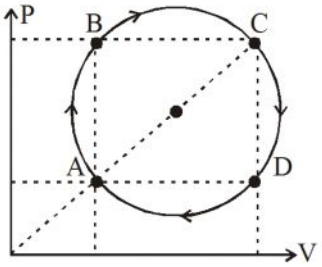
(A) have dispersion without average deviation.

(B) have deviation without dispersion.

(C) have both dispersion and average deviation.

(D) have neither dispersion nor average deviation.

4. A sample of ideal gas of some mass is taken through the cyclic process shown in the figure. The temperature of the gas at state A is $T_A = 200$ K, and at state C is $T_C = 1800$ K. Choose the correct option(s) :-



(A) $\frac{V_A}{V_C} = \frac{1}{3}$ (B) $\frac{V_A}{V_B} = \frac{1}{9}$ (C) $\frac{P_A V_A}{P_C V_C} = \frac{1}{3}$ (D) $\frac{P_A V_A}{P_C V_C} = \frac{1}{9}$

5. X-ray from a tube with a target A of atomic number Z shows strong K lines for target A and weak K lines for impurities. The wavelength of K_α lines is λ_z for target A and λ_1 and λ_2 for two impurities. If $\frac{\lambda_z}{\lambda_1} = 4$ and $\frac{\lambda_z}{\lambda_2} = \frac{1}{4}$. Screening constant of K_α lines to be unity. Select the correct statement(s).

(A) The atomic number of first impurity is $2Z - 1$.

(B) The atomic number of first impurity is $2Z + 1$.

(C) The atomic number of second impurity is $\frac{(Z+1)}{2}$.

(D) The atomic number of second impurity is $\frac{Z}{2} + 1$.

6. A man is sitting in a boat floating in water. There are heavy stones placed in the boat.

(A) When the man throws the stones in water, the level of boat goes down.

(B) When the man throws the stones in water, the level of boat rises up.

(C) When the man drinks some water, the level of boat goes down

(D) When the man drinks some water, the level of boat remains unchanged.

SECTION-I (ii) : (Maximum Marks: 12)

This section contains **FOUR (04)** questions.

Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**

For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

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7. A block of mass 1.0 kg is pulled up on a rough slope of inclination 37° with the horizontal. The coefficient of kinetic friction between the block and the slope is 0.25. The block moves with acceleration 2.0 m/s^2. After 5.0 s from the instant the block starts, which of the following statements is/are true ? (Power of any force is given by $P = \vec{F} \cdot \vec{V}$)

	List-I		List-II
(I)	Power delivered by the pulling force	(P)	20 W
(II)	Power delivered by the weight of the block	(Q)	–20 W
(III)	Power delivered by the friction between the slope and the block	(R)	–60 W
(IV)	Power delivered by the contact force between the slope and the block	(S)	100 W
		(T)	80 W

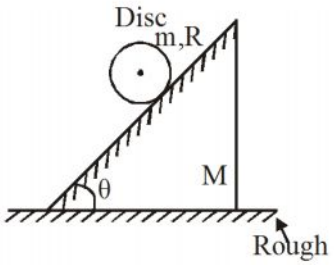
(A) I ⟶ P;II ⟶ R;III ⟶ Q;IV ⟶ S

(B) I ⟶ S;II ⟶ R;III ⟶ Q;IV ⟶ Q

(C) I ⟶ P;II ⟶ Q;III ⟶ R;IV ⟶ S

(D) I ⟶ Q;II ⟶ S;III ⟶ T;IV ⟶ P

8. A disc of mass m and radius R is performing pure rolling on the inclined surface of a wedge of mass M. Wedge is placed on a rough horizontal surface. If wedge does not move on the horizontal surface then match the followings.



	List-I		List-II
(I)	Frictional force acting on disc	(P)	$\frac{mg\sin\theta}{3}$
(II)	Vertical component of net frictional force acting on wedge	(Q)	$\frac{mg\sin^2\theta}{3}$
(III)	Normal reaction acting on disc	(R)	$mg\cos\theta$
(IV)	Normal reaction acting on wedge by the ground	(S)	$\left(m\cos^2\theta+\frac{m}{3}\sin^2\theta+M\right)g$
		(T)	$\left(m\sin^2\theta+\frac{m}{3}\cos^2\theta+M\right)g$

(A) I $\longrightarrow$ P;II $\longrightarrow$ Q;III $\longrightarrow$ S;IV $\longrightarrow$ T

(B) I $\longrightarrow$ P;II $\longrightarrow$ Q;III $\longrightarrow$ T;IV $\longrightarrow$ S

(C) I $\longrightarrow$ P;II $\longrightarrow$ Q;III $\longrightarrow$ R;IV $\longrightarrow$ S

(D) I $\longrightarrow$ S;II $\longrightarrow$ T;III $\longrightarrow$ P;IV $\longrightarrow$ Q

9. Match the following lists.

	List–I		List–II
(I)	Nearby objects appear moving past faster than distant objects.	(P)	Human eye sense the direction from which light enters it.
(II)	Extremely distant objects appear moving among with us	(Q)	It cannot resolve two points if they subtend an angle less than one minute.
(III)	Many people believe that they can read better in bright light than in dim light	(R)	Persistence of human eye is 0.1 s
(IV)	The colors of the surface of bodies that can be impregnated with water appear richer after moistening the surface.	(S)	Contraction of the pupil in bright light reduces spherical aberration.
		(T)	Due to moistening, surface irregularities are covered with water film, which reduces diffuse reflection of light

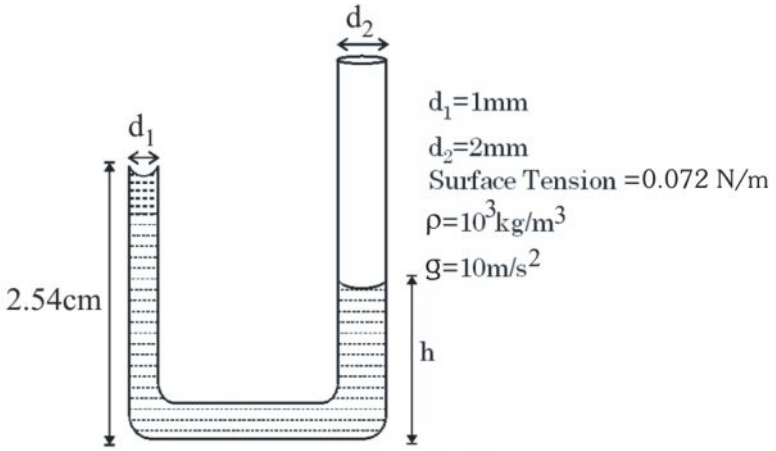
(A) I $\longrightarrow$ P;II $\longrightarrow$ Q;III $\longrightarrow$ R;IV $\longrightarrow$ S

(B) I $\longrightarrow$ P;II $\longrightarrow$ T;III $\longrightarrow$ R;IV $\longrightarrow$ Q

(C) I $\longrightarrow$ P;II $\longrightarrow$ S;III $\longrightarrow$ R;IV $\longrightarrow$ P

(D) I $\longrightarrow$ P;II $\longrightarrow$ P;III $\longrightarrow$ S;IV $\longrightarrow$ T

10.



The right limb is long enough and its radius is twice that of left limb. Water is slowly poured into the right limb. Considering water wets completely and the left limb is fully filled. List I corresponds to height of liquid in right limb and list II corresponds to radius of meniscus (R) and its shape in left limb.

	List-I		List-II
(P)	h = 1.1 cm	(1)	Concave upward
(Q)	h = 2 cm	(2)	Flat
(R)	h = 2.54 cm	(3)	Convex upward
(S)	h = 6.86 cm	(4)	R = 1mm
		(5)	R = 0.5 mm

(A) P ⟶ 1,4;Q ⟶ 1;R ⟶ 2;S ⟶ 3,4

(B) P ⟶ 1;Q ⟶ 1;R ⟶ 2;S ⟶ 3,4

(C) P ⟶ 1,5;Q ⟶ 1;R ⟶ 1,4;S ⟶ 3,5

(D) P ⟶ 1,5;Q ⟶ 1;R ⟶ 1,5;S ⟶ 3

SECTION-II : (Maximum Marks: 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
Answer to each question will be evaluated according to the following marking scheme:

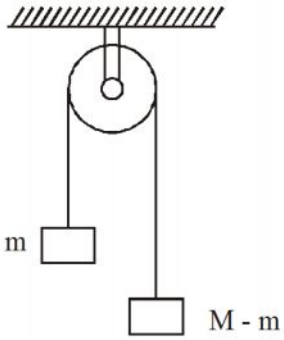
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

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1. In a long cylindrical metal wire of radius r and of resistivity ρ, a current of strength I flows in uniform distribution. The wire has constant surface temperature T_0. The temperature of the wire on its axis of symmetry is given by $T = T_o + \frac{I^2\rho}{\pi^2 r^2 K} \times \lambda$. Here K is thermal conductivity of wire. Find λ.

2. A certain amount of water is heated from 20ºC to 40ºC using an 800 Watt immersion heater. We expect to heat water in 210 seconds, but instead it takes 230 seconds. Determine the heat capacity of vessel in S.I. units. Ignore other heat losses.

3. A mass M is divided in two parts m and M−m. Now two parts are connected by an ideal string passing over a pulley as shown in diagram. Find value of m/M such that tension in string is maximum.



4. A cell of emf 1.5 V and internal resistance 0.5 Ω is connected to a nonlinear conductor in which current varies with voltage as $V^2 = I$. Calculate the current drawn from cell (in ampere).

5. A coil with inductance L and resistance R is connected to an alternating source. The capacity of a capacitor that can be connected in series later with the coil such that active power of circuit does not change is given by $C = \frac{n}{\omega^2 L}$. Find value of n.

6. Consider a spherical homogeneous cloud of mass M made by an explosion. Due to energy received from the explosion, it is expanding and the expansion is spherically symmetric. At an instant its radius is R_0 and the particles on the surface are moving away from its center with velocity V_0. Its radius when expansion ceases is given by $\frac{nGMR_0}{2GM - R_0V_0^2}$. Find n.

7. The electrostatic flux through a rectangular area of $\frac{7}{188}m^2$ is lying in a plane $2x + 3y + 6z = 10$ due to an electric field $(8\hat{i} + 6\hat{j} + 10\hat{k})$ V/m will be in (V-m)

8. A parallel beam of light of all wavelength greater than 3000 Å falls on a double slit in a Young's double slit experiment. It is observed that the wavelengths 3600Å and 6000Å are absent at a distance of 31.5 mm from the position of the central maximum, and the orders of the minima at this point for the two wavelengths differ by 7. If the distance between the slits and the screen is 1m, the separation between the two slits (in mm) is :

PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 24)

This section contains **SIX (06)** questions.

Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

For each question, choose the option(s) corresponding to (all) the correct answer(s)

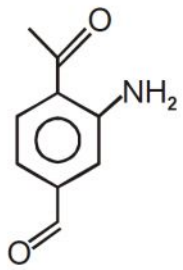
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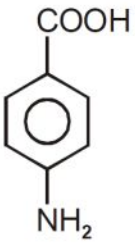
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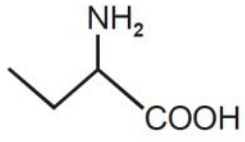
For Example : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

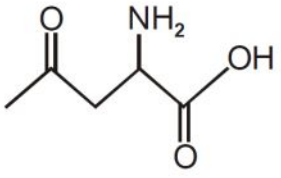
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1. A compound **'X'** evolves CO_2 when treated with $NaHCO_3$ but gives negative test with tollen's reagent and fehling's solution. On treatment with $CHCl_3$, KOH, a foul smelling organic compound is formed. If **'X'** do not give positive iodoform test, which of the following satisfy the conditions to be **'X'**.

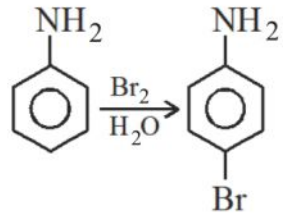
(A) 

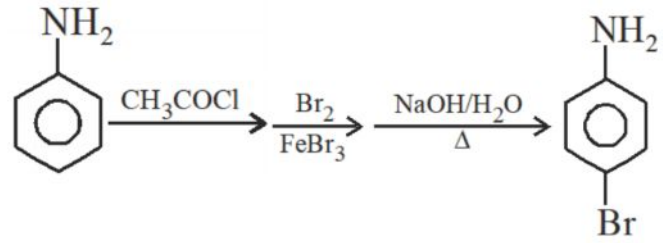
(B) 

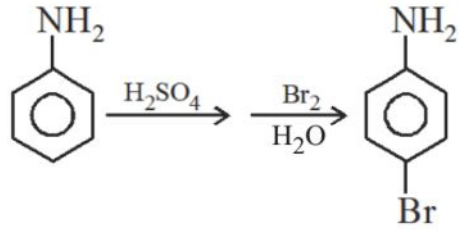
(C) 

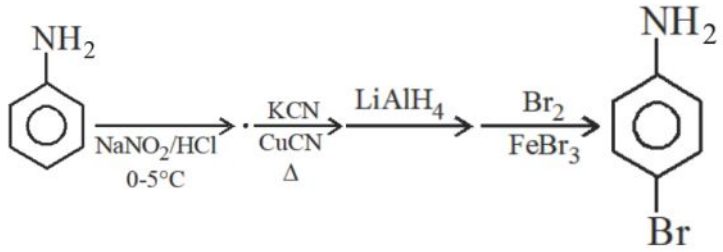
(D) 

2. Which of the following method(s) is/are not used to prepare p-bromo aniline as major product :

(A) 

(B) 

(C) 

(D) 

3. In which of the following process(es) electrons are removed from different type of orbitals?

(A) $O_2^- \rightarrow O_2^+$

(B) $Cu \longrightarrow Cu^{2+}$

(C) $N_2^+ \rightarrow N_2^-$

(D) $NO \longrightarrow NO^{2+}$

4. Which of the following statements are incorrect from following options?

(A) Zone refining is carried out in presence of noble gas environment.

(B) Fe is the 2nd most abundant metal in the earth crust.

(C) Sapphire consists of Al_2O_3 and Co (as impurity).

(D) Formula of Bauxite : $AlO_{3-2x}(OH)_x$, where $0 < x < 1$.

5. Which of following is/are incorrect statement(s) for a sample of ideal gas ?

(A) Temperature of gas remains constant during its adiabatic free expansion

(B) During compression of the gas at constant pressure, temperature of gas increases.

(C) Molar heat capacity of the gas is zero during an isothermal process

(D) At 300 K temperature molar internal energy change of $SO_2(g)$ is greater than molar internal energy change of $CO_2(g)$ when temperature of both gases is increased by 20 K.

6. Select the correct statement(s) about FCC(ABCABC...) structure,
[Given : a = Edge length of FCC unit cell]

(A) Distance between nearest octahedral void and tetrahedral void is $\frac{\sqrt{3}a}{4}$

(B) Distance between two nearest octahedral voids is $\frac{a}{\sqrt{2}}$

(C) Distance between two nearest tetrahedral voids is $\frac{\sqrt{3}a}{2}$

(D) Distance between close-packed layers A and B is $2r\sqrt{\frac{2}{3}}$ [r = radius of atom]

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

7.

List-I ((Compound)		List-II (Can be distinguish by)	
(P)	CH_3-CHO & $Ph-CHO$	(1)	Br_2 / H_2O test
(Q)	CH_3-CH_3 & $HC\equiv CH$	(2)	Ammonical Aq. NO_3
(R)	$CH_3-C\equiv CH$ & $CH_3-\overset{\overset{O}{\|\|}}{C}-CH_3$	(3)	Iodoform test
(S)	CH_3-CHO & $CH_3-\overset{\overset{O}{\|\|}}{C}-CH_3$	(4)	$NaHSO_4$ test
		(5)	Fehling test

(A) P ⟶ 3,5;Q ⟶ 1,2;R ⟶ 1,2,3,4;S ⟶ 2,5

(B) P ⟶ 1,2;Q ⟶ 2,3;R ⟶ 3,4;S ⟶ 4,5

(C) P ⟶ 3,4,5;Q ⟶ 2,3,4;R ⟶ 4,5;S ⟶ 1,2,3

(D) P ⟶ 3,4;Q ⟶ 2,4;R ⟶ 1,2;S ⟶ 4,5

8. List-I contains co-ordination compounds and List-II contains their properties:

List-I	List-II
(P) $[NiCl_4]^{2-}$	(1) Diamagnetic species
(Q) $[Cr(H_2O)_3F_3]$	(2) Low spin octahedral complex
(R) $[PtCl_4]^{2-}$	(3) Contains unpaired electrons
(S) $[Co(H_2O)_6]^{3+}$	(4) d^8 configuration of metal ion
	(5) Exhibits geometrical isomerism

[Given atomic no. Ni = 28, Cr = 24, Pt = 78, Co = 27]

Match each species in List-I with their properties in List-II and choose the correct option:

(A) P ⟶ 1,2;Q ⟶ 3,4;R ⟶ 3,5;S ⟶ 1,4

(B) P ⟶ 3,4;Q ⟶ 2,5;R ⟶ 1,4;S ⟶ 1,2

(C) P ⟶ 4,5;Q ⟶ 1,2;R ⟶ 2,3;S ⟶ 3,5

(D) P ⟶ 3,4;Q ⟶ 3,5;R ⟶ 1,4;S ⟶ 1,2

9. Boyle's temperature (T_B): T_B of CO_2 = 350°C, T_B of CH_4 = 60°C and T_C of H_2 = −240ºC

List-I		List-II	
(P)	CO_2 gas in low pressure region & temperature 573 K.	(1)	PV = nRT
(Q)	CO_2 gas in very high pressure region & temperature 623 K	(2)	Gas is more compressible with respect to an ideal gas.
(R)	CH_4 gas at 333 K & in low pressure region.	(3)	Gas cannot be liquefied at any pressure at given temperature.
(S)	H_2 gas in low pressure region & temperature 20 K.	(4)	Molar volume of gas is lesser than that of ideal gas at same pressure and temperature.
		(5)	Repulsive forces dominate

(A) P ⟶ 2,3;Q ⟶ 1,2;R ⟶ 1,2;S ⟶ 2,4

(B) P ⟶ 2,3,4,5;Q ⟶ 3,5;R ⟶ 1,3;S ⟶ 2,4

(C) P ⟶ 2,3,4;Q ⟶ 3,5;R ⟶ 1,3;S ⟶ 1,2,4

(D) P ⟶ 2,3,4;Q ⟶ 3,5;R ⟶ 1,3;S ⟶ 2,4

10. Match the following

List - I		List - II	
(P)	A saturated aqueous solution of $Mg(OH)_2$ $[K_{sp} = 4\times10^{-12}]$	(1)	pH = 7
(Q)	300 ml of 0.1 M BOH +100 ml. of 0.1M HCl $[K_b(BOH) = 10^{-4}]$	(2)	$pH = \frac{pKa_2 + pKa_3}{2}$
(R)	300 ml. of 0.1 M BOH+300 ml. of 0.1 M HA $[K_a = K_b = 10^{-4}]$	(3)	pH= 10 + log 2
(S)	0.01 M aqueous solution Na_2HPO_4 [Given that: $pK_{a_1}(H_3PO_4) = 4.3$; $pK_{a_2}(H_3PO_4) = 8.102$; $pK_{a_3}(H_3PO_4) = 12.5$;	(4)	$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a - \frac{1}{2}pK_b$
		(5)	Buffer solution

(A) P ⟶ 3;Q ⟶ 3;R ⟶ 2;S ⟶ 3

(B) P ⟶ 1;Q ⟶ 3;R ⟶ 1;S ⟶ 4

(C) P ⟶ 3;Q ⟶ 1;R ⟶ 4;S ⟶ 3

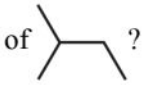
(D) P ⟶ 3;Q ⟶ 3;R ⟶ 5;S ⟶ 2

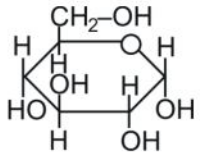
SECTION-II : (Maximum Marks: 24)

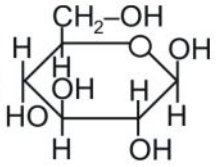
This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
Answer to each question will be evaluated according to the following marking scheme:

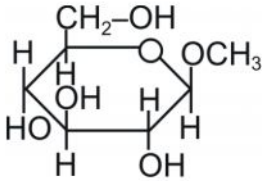
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

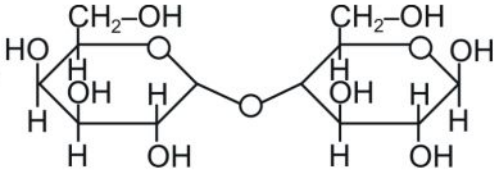
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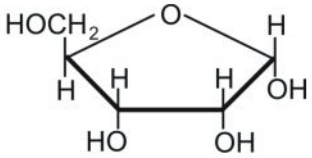
1. Total number of fractions obtained on fractional distillation of all monochloro derivatives product(s) of  ?

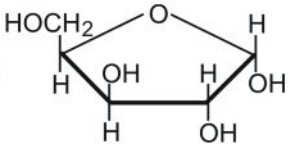
2. (A) 

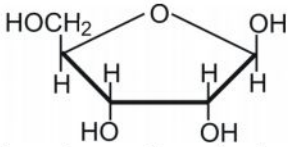
(B) 

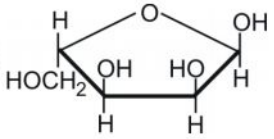
(C) 

(D) 

(E) 

(F) 

(G) 

(H) 

Calculate the number of reducing sugar = x
Calculate the number of sugar in which molecular weight increases by 210, whenever it undergo acetylation reaction with acetic anhydride = y
So x – y =

3. Organic compound [M] has the following data :
(i) Molecular weight is 149, it contains only 3 different elements.
(ii) It can rotate plane polarized light (chiral carbon).
(iii) [M] on oxidation with hot alkaline $KMnO_4$ produces benzoic acid.
(iv) [M] can be prepared by treating carbonyl compound with methyl amine followed by $NaBH_4$.
Write the number of possible structures for [M] (excluding stereo isomers).

4. A mixture of Potash alum, Chrome alum and Mohr's salt is taken in water and H_2O_2 is added in strongly alkaline medium and finally filtered.

If atomic mass/molecular mass of cation in residue is "a" and the number of type of ions from the salt mixture passed into the filtrate is 'b'. Then find the value of a/b.

(Atomic mass : Cr = 52 , Fe = 56 , S = 32 , O = 16 , N = 14 , C = 12 , B = 11 , H = 1 , K^+ = 39 , Na^+ = 23 , Ni = 58 , Cu = 29 , Zn = 65 , Cl = 35.5 , Br = 80 , F = 19)

5. (a) Fullerene has no dangling bonds.

(b) Ni + excess NH_3 solution $\longrightarrow$ deep blue solution

(c) PH_3 can be absorbed by $HgCl_2$(aq.)

(d) Bleaching powder available in market is completely soluble in water.

(e) Na can be safely stored in ethanol.

(f) $Na_2S_2O_3$ + Cl_2-water (excess) produces $NaHSO_4$ and HCl.

(g) Higher concentration of SO_2 damages the leaves of plant and retard the rate of photosynthesis.

If x = number of correct statements.

y = number of incorrect statements.

then find the value of x/y.

6. Let 'A' and 'B' are two completely miscible liquids. P_A^o = 90 torr and P_B^o = 40 torr. In a flask 8 moles of 'A' is mixed with 4 moles of 'B'. However as soon as B is added 'A' starts dissociating as two different liquids 'C' and 'D' as A $\longrightarrow$ C + D.

Both 'C' and 'D' are also completely miscible with each other and with 'A' and 'B'. Given P_C^o = 50 torr and P_D^o = 40 torr.

Assume dissociation of 'A' follows 1st order kinetics with half life equal to 50 min. After 100 min 2 moles of a non-volatile non-electrolyte solute 'S' is added in solution which immediately arrests the dissociation of A completely.

Calculate final vapour pressure of the solution in torr.

7. Fixed amount of an ideal gas contained in a sealed rigid vessel of volume 24.6 litres at 1. 0 bar is heated reversibly from 27ºC to 127ºC. The entropy of gas is S = 10 + 10^{-2} T (J/K). Then what is the enthalpy change (in J) during the process.

8. NO_3^- $\rightarrow$ NO_2(acid medium), E^o=0.790V

NO_3^- $\rightarrow$ NH_3OH^+(acid medium) E^o=0.731V.

At what pH. the above two electrodes will have same electrode potential value?

Assume that activity of all other species to be unity except [H^+] ions ? $\left(\frac{2.303\,RT}{F} = 0.059\right)$

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 24)

This section contains **SIX (06)** questions.

Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

For each question, choose the option(s) corresponding to (all) the correct answer(s)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

For Example : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

1. Let the equation $ax^2 - bx + c = 0$ has 2 distinct roots in the interval (0,1) where $a,b,c \in N$. If $\lambda < \log_5(abc)$ for all choices of natural numbers a,b,c then non-negative integral values of λ can be :

(A) 0 (B) 1

(C) 2 (D) 3

2. Event 'A' is independent of event B, $B \cup C$ and $B \cap C$. If $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(C) = \frac{1}{4}$. Then : (Where P(x) denotes probability of event x)

(A) $P\left(\frac{A}{C}\right) = \frac{1}{2}$

(B) $P\left(\frac{\bar{B} \cup \bar{C}}{A}\right) = \frac{11}{12}$ (where B and C independent events)

(C) $P\left(\frac{\bar{A}}{\overline{B \cap C}}\right) = \frac{1}{2}$

(D) A and C are not independent events

3. An ellipse with eccentricity $\frac{1}{2}$ passes through P(3,4) whose nearer focus is S(0,0) and equation of tangent at P on ellipse is $3x + 4y - 25 = 0$. If a chord through S parallel to tangent at P intersects the ellipse at A and B then :

(A) length of AB is 15

(B) length of latus rectum of ellipse is 15

(C) focal length of ellipse is 10

(D) centre of ellipse is $(-3,-4)$

4. Let $< T_n >$ be a sequence such that $T_n^3 + 2T_n = T_{n+1}$ $\forall$ $n \in N$, and $T_1 = 1$, then :

(A) $\sum_{n=1}^{100} \left(T_n^3 + T_n + 1\right) = T_{101} + 99$

(B) $\sum_{n=1}^{100} \left(T_n^3 + T_n + 1\right) = T_{101} + 100$

(C) $\prod_{n=1}^{100} \left(T_n^2 + 2\right) = T_{100}$

(D) $\prod_{n=1}^{100} \left(\frac{T_{n+1}}{T_n} - T_n^2\right) = 2^{100}$

5. Let $f : (0, \infty) \longrightarrow R$ be a differentiable function satisfying the equation $f(xy) = e^{xy-x-y}(e^y f(x) + e^x f(y))$ $\forall$ $x, y > 0$. If $f'(1) = e$, then

(Note : where 'e' is nepier's constant)

(A) $\lim_{x \to e}\left[\frac{f(x) - e^x}{x - e}\right] = e^{(e-1)}$

(B) number of roots of the equation $f(x) = xe^x$ in $(0,\infty)$ is 2.

(C) $\int_1^e f(x)\,dx < e^e(e-1)$

(D) $f(x)$ is strictly increasing function in $(0,\infty)$.

6. If three planes $P_1 \equiv 2x + y + z - 1 = 0$, $P_2 \equiv x - y + z - 2 = 0$ and $P_3 \equiv \alpha x - y + 3z - 5 = 0$ intersects each other at point P on XOY plane and at point Q on YOZ plane, where O is the origin then identify the correct statement(s) ?

(A) The value of α is 4.

(B) Straight line perpendicular to plane P_3 and passing through P is $\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$.

(C) The length of projection of $\overrightarrow{PQ}$ on x-axis is 1.

(D) Centroid of the triangle OPQ is $\left(\frac{1}{3}, \frac{-1}{2}, \frac{1}{2}\right)$

SECTION-I (ii) : (Maximum Marks: 12)

This section contains **FOUR (04)** questions.

Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**

For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

7. Match the following List-I with List-II

	List-I		**List-II**
(I)	The number of all triples (a,b,c) such that all three term a,b,c are in harmonic progression in which a = 20 and b divides c (where a,b,c are strictly increasing positive integers)	(P)	66
(II)	The number of solutions of m + n + p = 10, (m,n,p are non-negative integers)	(Q)	20
(III)	The number of ordered pair (a,b) satisfying the equation ab + b – a + 1 = 0 are where a,b ∈ I	(R)	4
(IV)	The number of ordered pairs (x,y) of real numbers such that $(x + iy)^{20} = x - iy$ is (where $i = \sqrt{-1}$)	(S)	5
		(T)	22

(A) I ⟶ S;II ⟶ P;III ⟶ R;IV ⟶ T

(B) I ⟶ T;II ⟶ Q;III ⟶ S;IV ⟶ P

(C) I ⟶ Q;II ⟶ P;III ⟶ R;IV ⟶ T

(D) I ⟶ R;II ⟶ S;III ⟶ Q;IV ⟶ P

8. Match the following List-I with List-II

	List-I		**List-II**
(I)	Let $f(x) = x^4 + ax^3 + bx^2 + cx + d$ (where a,b,c,d are real coefficient) and $f(x) = 0$ has real roots, If $\|f(i)\| = 1$ and $a = b = c = d = k$ then k can be (where $i = \sqrt{-1}$)	(P)	0
(II)	If $x^5 = 1$ $(x \neq 1)$ then $\frac{x}{1+x^2} + \frac{x^2}{1+x^4} + \frac{x^3}{1+x} + \frac{x^4}{1+x^3}$ equals	(Q)	1
(III)	Let $f(x) = \ell n\left[\cos\|x\| + \frac{1}{2}\right]$ (where [.] denotes the greatest integer function), then $\int_{x_1}^{x_2}\left(\lim_{n\to\infty}\left(\frac{(f(x))^n}{x^2+\tan^2 x}\right)\right)dx$ is (where $x_1x_2 \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$)	(R)	2
(IV)	Let ABCD be a cyclic quadrilateral inscribed in a circle of radius 2, such that $BD = 2\sqrt{3}$, $AB = 1$. If $\angle BAD$ is acute and $AD = a + b\sqrt{5}$, then the value of $a + b$ is equal to	(S)	3
		(T)	4

(A) I $\longrightarrow$ Q;II $\longrightarrow$ P;III $\longrightarrow$ R;IV $\longrightarrow$ T

(B) I $\longrightarrow$ P;II $\longrightarrow$ R;III $\longrightarrow$ P;IV $\longrightarrow$ R

(C) I $\longrightarrow$ T;II $\longrightarrow$ R;III $\longrightarrow$ Q;IV $\longrightarrow$ S

(D) I $\longrightarrow$ S;II $\longrightarrow$ P;III $\longrightarrow$ R;IV $\longrightarrow$ Q

9. Match the following List-I with List-II

	List-I		List-II
(I)	If $f(x) + f''(x) = -x\, g(x) f'(x)$ and $g(x) > 0\ x \in R$ then $(f(x))^2 + [f'(x)]^2$ has k as its maxima. Then k is	(P)	0
(II)	If $f^2(x) + f(x) = g(x)$ and g(x) is always increasing then the minimum value that f(x) can attain so that f(x) also increasing is k. Then (2(k + 1)) is	(Q)	1
(III)	If $g(x) = f(x) \int_0^x f(t)\,dt$ is non-increasing function and f(x) is continuous in R then f(1) is	(R)	2
(IV)	If f(x) is differentiable function and $f(x) = x^2 + \int_0^x e^{-t} f(x-t)\,dt$ then $f(-1) = \frac{k}{3}$ then k is	(S)	−1
		(T)	3

(A) I ⟶ P;II ⟶ Q;III ⟶ P;IV ⟶ R

(B) I ⟶ Q;II ⟶ T;III ⟶ P;IV ⟶ R

(C) I ⟶ S;II ⟶ Q;III ⟶ T;IV ⟶ S

(D) I ⟶ R;II ⟶ T;III ⟶ P;IV ⟶ Q

10. Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where a,b > 0 be a hyperbola in the xy-plane whose centre is C, one of the foci is F_1. Foot of perpendicular from F_1 upon line L : bx − ay = 0 is P. Triangle CF_1P is isosceles triangle whose area is 8 sq. units.

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(I)	The length of transverse axis of H is	(P)	$4\sqrt{2}$
(II)	The eccentricity of H is	(Q)	1
(III)	The slope of L is	(R)	4
(IV)	The distance between directrices of H is	(S)	$\sqrt{2}$
		(T)	8

(A) I ⟶ S;II ⟶ T;III ⟶ Q;IV ⟶ P

(B) I ⟶ T;II ⟶ S;III ⟶ Q;IV ⟶ P

(C) I ⟶ T;II ⟶ S;III ⟶ Q;IV ⟶ R

(D) I ⟶ T;II ⟶ R;III ⟶ Q;IV ⟶ P

SECTION-II : (Maximum Marks: 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

---

1. Suppose f and g are differentiable function such that $xg(f(x))f'(g(x))g'(x) = f(g(x))g'(f(x))f'(x)$ for all real x. Also f is non negative and g is positive. If $\int_0^a f(g(x))\,dx = \frac{1}{2} - \frac{e^{-2a}}{2}$ for all reals a and g(f(0)) = 1 then the value of g(f(4)) is equal to $e^{-\lambda}$ where $\lambda \in N$, then the value of $\frac{\lambda}{10}$ is

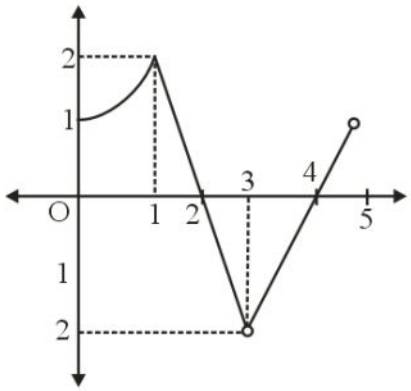
Note : (where 'e' is napier's constant)

2. Let z satisfies $z\bar{z} + (-4+5i)\bar{z} + (-4-5i)z - 40 = 0$. If $a = \max.|z+2-3i|$ and $b = \min|z+2-3i|$, then value of $\left(\frac{a-b}{8}\right)$ is ?

3. Let $f(1) + g(1) = 9e$; $f(x) = -x^2g'(x)$; $g(x) = -x^2f'(x)$. If $\int_1^4 \frac{f(x)+g(x)}{x^2}dx = k\left(e - e^{\frac{1}{4}}\right)$, then the value of $\frac{k}{4}$ is

(Note : where e is napier's constant)

4. Graph of a function y = f(x) is shown as



If g(x) = |f(|x|)|, then find number of solution(s) of the equation

$g(g(x)) = \text{sgn}(x^2 - (k+1)x + (k^2+1))$, $k \in R$.

[Note : sgn(y) denotes the signum function of y.]

5. Let $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ and $P = \begin{bmatrix} 7 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 5 \end{bmatrix}$. Consider $A = P^{-1}DP$ then $\frac{\det.\left(A^2 + A\right)}{10}$ is

6. Let $A_r, r = 1,2,.....29$ be arithmetic means between 303 and -57 where $A_r > A_{r+1}\ \forall\ r = 1,2,....28$. If S be the sum of these means, then the value of $\left(\frac{S}{(A_{14} - 12)\,|A_r|_{min}}\right)$

[Note : $|A_r|_{min}$. denotes the minimum value of $|A_r|$]

7. If $\left(x - 2 + \frac{1}{x}\right)^{30} = n_0x^{30} + n_1x^{29} + \ldots + n_{29}x + n_{30} + n_{31}x^{-1} + \ldots + n_{60}x^{-30}$ and $C = n_0 + n_1 + n_2 + \ldots + n_{60}$. Find the value of a if $n_{30} - C = \binom{60}{a}$.

[Note : $\binom{n}{r}$ denotes ${}^nC_r$.]

8. Let $E - ABCD$ be a pyramid on square base ABCD where A is the origin and B and D are lying on positive x-axis and y-axis respectively. If E is (0,2,3) and $\overrightarrow{DE}.\left(\hat{i} + \hat{j}\right) = \vec{0}$, then volume of the tetrahedron ABDE is

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work



English (1001CJA101021230068)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
19-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $\mathbf{g = 10\ m/s^2}$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

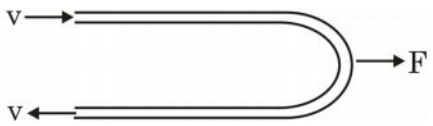
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

1. A horizontal tube of uniform cross-sectional area A is bent in the form of U as shown in figure. If the liquid of density ρ enters and leaves the tube with velocity v, then the external force F required to hold the bend stationary is :-



(A) $F = 0$

(B) ρAv^2

(C) $2\rho Av^2$

(D) None of these

2. The activity of a radioactive sample goes down to about 6% in a time of 2 hours. The half life of sample in minutes is about :-

(A) 30

(B) 15

(C) 60

(D) 120

3. A planet of mass m is revolving in an elliptical orbit about the sun with an orbital period T. If M_{sun} be the mass of sun and A be the area of orbit, then the angular momentum of the planet is :-

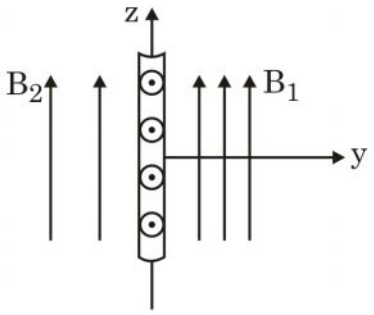
(A) $\frac{mA}{2T}$

(B) $\frac{mA}{4T}$

(C) $\frac{2mA}{T}$

(D) $\frac{4mA}{T}$

4. Figure shows a thin metal sheet in the plane y = 0, for which the current of constant density flows in the positive x-direction. It is in a constant homogeneous magnetic field of value $B_0\hat{k}$. As a result of superposition of magnetic field in region y > 0, the induction field $B_1\hat{k}$ and in y < 0 is $B_2\hat{k}$ where $B_1 > B_2$. Specify the correct statement :-



(A) $B_0 = \frac{(B_1 - B_2)}{2}$

(B) $B_0 = \frac{(B_1 + B_2)}{2}$

(C) $B_0 = B_1 + B_2$

(D) $B_0 = B_1 - B_2$

SECTION-I (ii) : (Maximum Marks: 12)

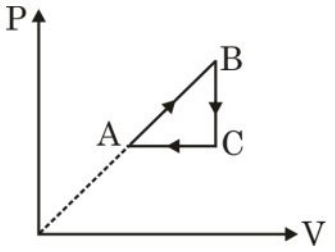
- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. Consider a thremodynamics cycle in a PV diagram shown in the figure performed by one mole of a monoatomic gas. The temperature at A is T_0 and volumes are related as $V_B = V_C = 2V_A$. Choose the correct options :-



(A) The maximum temperature during the cycle is $4T_0$

(B) Net work done by the gas during the cycle is $0.5\ RT_0$

(C) The heat capacity of the process AB is 2R.

(D) The efficiency of the cycle is 8.33%

6. The plates of a parallel plate capacitor with no dielectric are connected to a voltage source. Now a dielectric of dielectric constant K is inserted to fill the whole space between the plates with voltage source remaining connected to the capacitor. Select the correct statement(s) :-

(A) The energy stored in the capacitor will become K-times the initial energy.

(B) The electric field inside the capacitor will decrease K-times the initial field.

(C) The force of attraction between the plates will become K^2-times the initial force.

(D) The charge on the capacitor will become K-times the initial charge.

7. In a resonance tube experiment to determine speed of sound, air column in the pipe is made to resonate with a given fork of frequency 480 Hz. At room temperature T using a resonance tube a student finds two resonance position 16.0 cm and 51.0 cm, length of the air column for the first and the second resonance respectively. It is known that speed of the sound at 273 K is equal to 330 m/s. Then,

(A) percentage error in the measurement of speed of sound is $\frac{4}{7}\%$

(B) diameter of the tube is 5 cm.

(C) temperature of the room T is nearly 283 K.

(D) percentage error in the determination of T is $\frac{8}{7}\%$

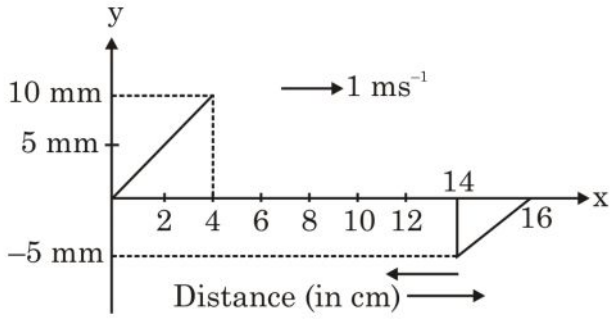
SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Two wave pulses are travelling in opposite direction with speed 1 ms^{-1}. Figure shows the shape of pulse at t = 0. Based on the above facts, answer the following questions.

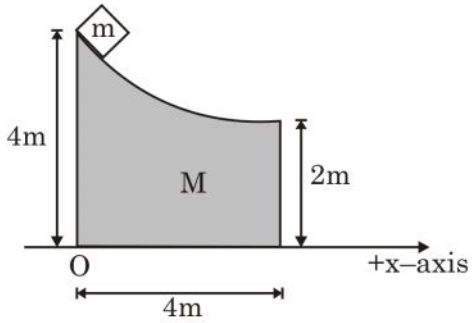


1. The speed of particle (in m/s) at x = 2 cm and t = 0 is :-

2. The displacement of particle (in mm) at x = 8 cm and t = 0.06 sec is

Paragraph for Questions 3 and 4

A small block of mass m = 1kg is placed over a wedge of mass M = 4kg as shown in figure. Mass m is released from rest. All surfaces are smooth and the origin O is fixed on the ground shown. Based on above information, answer the following questions.



3. Final velocity of the wedge is (in m/s)

4. The block strikes the x-axis at (in m)

SECTION-II (ii) : (Maximum Marks: 24)

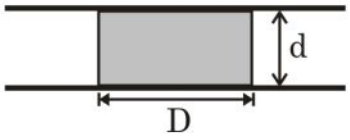
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

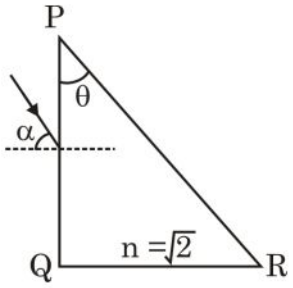
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5. In LCR circuit current resonant frequency is 600 Hz and half power points are at 650 and 550 Hz. The quality factor is

6. In a zero gravity region, a drop of a liquid of surface tension σ assumes a cylindrical shape of diameter D between two parallel glass plates that are a distance 'd' apart. If curved surface of the drop is at right angles to the plates as shown in the figure and force exerted by the drop on a plate is $\frac{\pi\sigma D}{n}$, then find 'n'



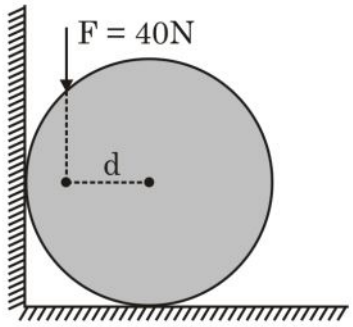
7. A parallel beam of light is incident from air at an angle 'α' on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45°. The angle 'θ' of the prism is 'n' times 3°. Find 'n'.



8. A wire having a length L and cross-sectional area A is suspended at one of its ends from a ceiling. Density and Young's modulus of material of the wire are ρ and Y, respectively. Its strain energy due to its own weight is $\frac{\rho^2 g^2 AL^3}{pY}$. Calculate the value of p.

9. In a hypothetical system a particle of mass m and charge $-3q$ is moving around a very heavy particle having charge q. Assuming Bohr's model to be applicable to this system, the orbital velocity of mass m when it is nearest to heavy particle is $\frac{3q^2}{x\varepsilon_0 h}$. Find the value of x.

10. A cylinder of radius r = 0.1 m and mass M = 2 kg is placed such that it is in contact simultaneously with a vertical and a horizontal surface as shown in figure. The coefficient of static friction is $\mu = \left(\frac{1}{3}\right)$ for both the surfaces. Find the distance d, in cm, from the centre of the cylinder at which a force F = 40 N should be applied so that the cylinder just starts rotating in the anticlockwise direction. Take g = 10 ms^{-2}.



PART-2 : CHEMISTRY

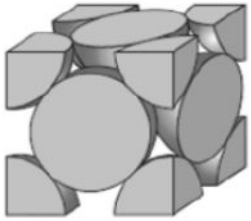
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

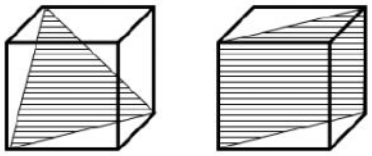
Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

1. Consider the following unit cell of a metal:



Also consider the following shaded planes P and Q representing two 2D lattices.



P Q

The ratio of packing fraction of P to Q is

(A) 1.00

(B) 1.63

(C) 0.55

(D) 0.74

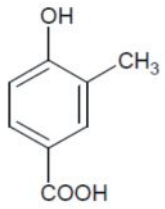
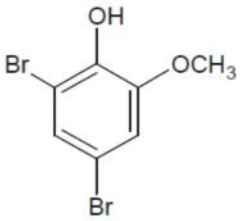
2. Vanillin (L), $C_8H_8O_3$, is isolated from vanilla beans. It give an intense colour with $FeCl_3$ and positive Tollen's test. It is not steam-distilled and does not react with HCl. It goes through following steps :

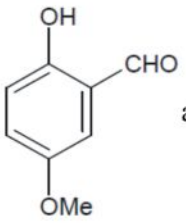
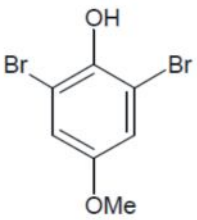
$$L \xrightarrow[\text{1. equivalent of } CH_3I]{OH^-} M,(C_9H_{10}O_3) \xrightarrow{KMnO_4} N,(C_9H_{10}O_4),(HCO_3^- \text{soluble})$$

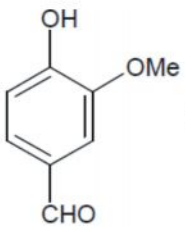
$$N \xrightarrow{conc.HI\ \Delta} \text{3,4-dihydroxybenzoic acid}$$

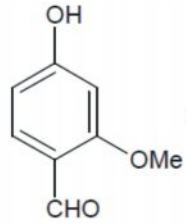
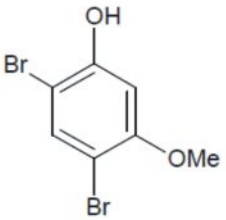
$$L \xrightarrow{[Ag(NH_3)_2]} O \xrightarrow{2.\ Br_2} P,(C_7H_6Br_2O_2)$$

Which of the following is correct statement about various products?

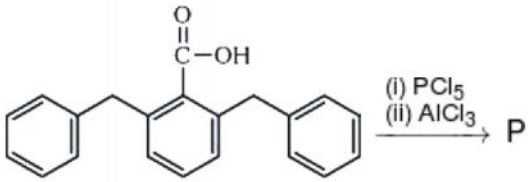
(A) L is  and P is 

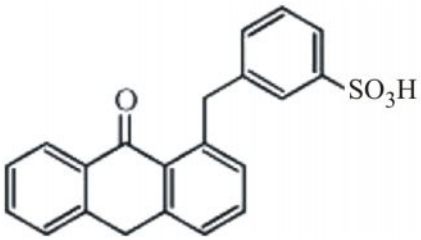
(B) L is  and P is 

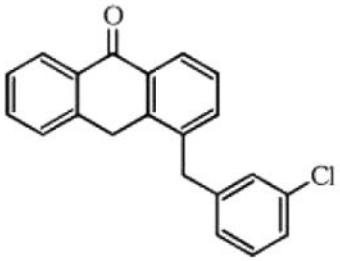
(C) L is  and P is 

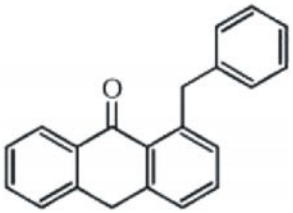
(D) L is  and P is 

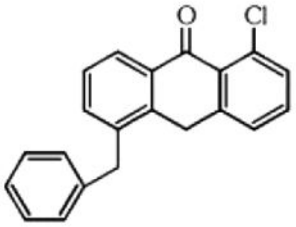
3. Predict the major product (P) in the following reaction :



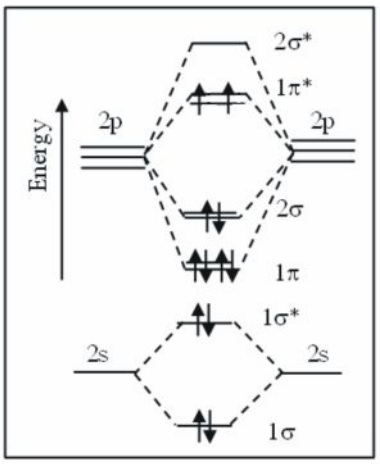
(A) 

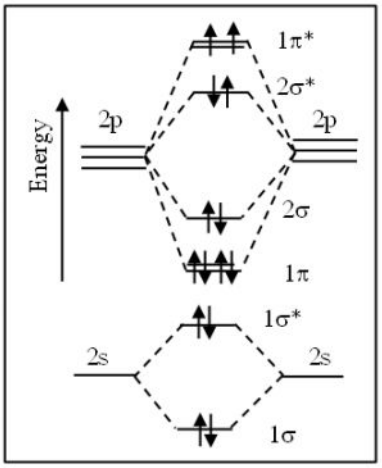
(B) 

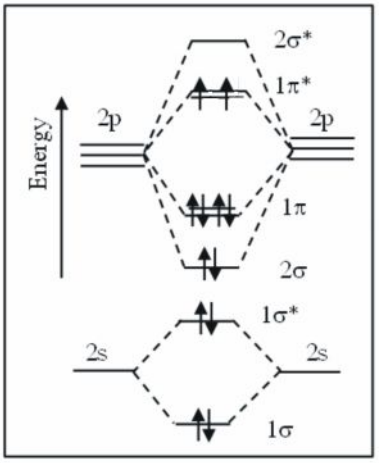
(C) 

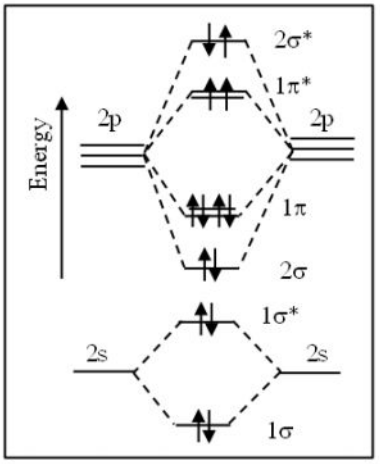
(D) 

4. The correct molecular orbital diagram for O_2 molecule in the ground state is

(A) 

(B) 

(C) 

(D) 

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. Consider one mole of a real gas A, with equation of state $P(V - b) = RT$ where 'b' is a constant. Which of the following is/are correct statement(s) about A?
(P = pressure, V=volume, T-temperature in Kelvin, R = 8.314 J K mol^{-1})

(A) Density of A is less than ideal gas at all pressure values.

(B) Potential energy of A, due intermolecular forces, is either greater than (or) equal to that of ideal gas

(C) If 5J of heat is given to A, and 2J work is done by the A then, its internal energy increases by 3J, which is true for ideal gas also

(D) Compressibility factor (Z) of A is directly proportional to 'P'.

6. When copper is reacted with dil, HNO_3 a gas is liberated. Which of the following are true for the gas?

(A) Has 11 valence e's, where extra electron occupies an antibonding π *2p orbital

(B) An acid anhydride

(C) Paramagnetic

(D) Neutral oxide

7. The complex(es), which can exhibit the type of isomerism shown by $[Co(en)_3]Cl_3$, is(are)
[en = $H_2NCH_2CH_2NH_2$]

(A) $[Pt(en)(SCN)_2]$

(B) $[Zn(gly)_2]$

(C) $[Co(NH_3)_4Cl_2]^+$

(D) $[Cr(en)_2(H_2O)(SO_4)]^+$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
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- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
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Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

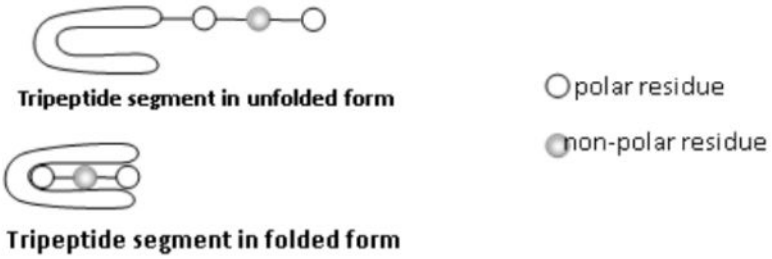
Consider the following process of folding/unfolding of a tripeptide segment in a large protein. The free energy change (ΔG) will depend on the interaction of the unfolded tripeptide with the solvent (water) and with the rest of the protein in the folded state (see below). Assume that the tripeptide is made up of one non-polar (hydrophobic, shaded) and two polar (hydrophilic; unshaded) residues. Use the following ΔG data of formation of interaction between residue with water or protein :

i) a non-polar residue and the solvent (water) $\Delta G = +8$ kJ mol^{-1}

ii) a non-polar residue and the rest of the protein $\Delta G = -4$ kJ mol^{-1}

iii) a polar residue and the solvent (water) $\Delta G = -16$ kJ mol^{-1}

iv) a polar residue and the rest of the protein $\Delta G = -14$ kJ mol^{-1}

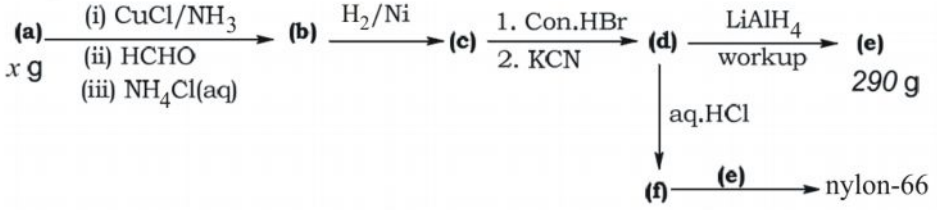


1. What is the magnitude of ΔG for the folding of the tripeptide segment in KJ/mol?

2. What is the magnitude of ΔG for the folding of the tripeptide segment if all the three residues are polar in KJ/mol?

Paragraph for Questions 3 and 4

Given below is synthesis of nylon-66. Assume 100 % yield in every step and none of the organic compounds involved are in excess.



3. The value of x is _____ .

4. The value of y is _____ .

SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

---

5. A solution of 0.1 M CH_3COOH is placed between parallel electrodes of cross-section area 4 cm^2, separated by 2 cm. For this solution, resistance measured is 100 Ω. The elevation in boiling point of the 0.1 M CH_3COOH solution is $x \times 10^{-5}$. The value of x is (use following information)
$K_b = 0.5K$ kg / mol; $\Lambda_m^{\infty}\left(H^+\right) = 300$ S cm^2 $mole^{-1}$; $\Lambda_m^{\infty}\left(CH_3COO^-\right) = 100$ S cm^2 $mole^{-1}$

6. $H_3C-\overset{\overset{O}{\|}}{C}-CH_3 + I_2 \xrightarrow{NaOH}$ Iodoform reaction

How many molecules of I_2 are required for 3 molecules of CH_3COCH_3?

7. Find total number of inner orbital diamagnetic complexes :
$[Fe(CN)_6]^{3-}$, $[Co(H_2O)_6]^{2+}$, $[Cr(en)_3]^{3+}$, $[Fe(CN)_6]^{4-}$, $[Co(H_2O)_6]^{3+}$, $[Cr(CN)_6]^{3-}$, $[PtCl_6]^{2-}$, $[Ni(H_2O)_6]^{2+}$, $[Cr(CO)_6]$

8. Consider following statements :

(1) $CO_2(g) + C(s) \longrightarrow 2CO(g)$, it is endothermic reaction,

(2) In copper extraction from $CuFeS_2$, $CaSiO_3$ is main slag formed.

(3) Brass is an alloy consisting of Zn metal.

(4) $Cu + H_2SO_4(conc.) \longrightarrow CuSO_4 + H_2$.

(5) In Hall-Heroult process, anode is corroded forming CO_2 gas only.

(6) Due to non-metallic impurtites melting point of iron is raised.

(7) In extraction of copper from copper pyrite, roasting process is employed.

How many of the following are correct statements.

9. Consider the following precipitate/s

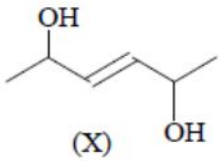
PbS , CuS , HgS , CdS , As_2S_3, As_2S_5, SnS, SnS_2, Sb_2S_3, Sb_2S_5

x = Total number of yellow ppt/s which soluble in yellow ammonium sulphide

y = Total number of black ppt/s which insoluble in hot & dil. HNO_3

Find the value of $\frac{x}{y}$.

10.



(X)

Identify total number of stereoisomer's if double bond is present in cis configuration in given compound (X)

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

1. Consider $f : [0, \infty) \longrightarrow [0, \infty)$ is bijective function and it satisfies $f(x) \cdot e^{f(x)} = x$, then the value of $\lim\limits_{x\to\infty} \dfrac{f(x)}{\ln x}$ is

(A) 0

(B) 1

(C) e

(D) Does not exist

2. Real number x, y, z satisfies

$x + xy + xyz = 1$

$y + yz + xyz = 2$

$z + xz + xyz = 4$

If the largest possible value of xyz is $\dfrac{a + \sqrt{b}}{c}$, where b & c are prime numbers, then (a + b + c) equals

(A) 20

(B) 22

(C) 24

(D) 26

3. If y is implicit differentiable function of x such that $y(x+y)^2 = x$, then $\int \frac{dx}{x+3y}$ is equal to

(A) $\frac{1}{2}\ln\left((x+y)^2+1\right)+C$

(B) $\ln\left((x+y)^2+1\right)+C$

(C) $\frac{1}{2}\ln\left((x-y)^2+1\right)+C$

(D) $\ln\left((x-y)^2+1\right)+C$

Where C is constant of integration

4. Let $A = \{1,2,3,4,5,6,7,8\}$ be a set. Let $f : A \longrightarrow A$ be a function such that $f(x) \neq x \ \forall \ x \in A$, $f(x)$ is even whenever x is even and $f(x)$ is odd whenever x is odd. Then total number of onto functions satisfying these conditions will be

(A) 36

(B) 49

(C) 64

(D) 81

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. Let $f : R \to R$, $g : R \to R$ and $h : R \to R$ be differentiable function such that $f(x) = x^3 + 5x + 1$, $g(f(x)) = x$ and $h(g(g(x))) = x \ \forall \ x \in R$ then

(A) $h(0) = 7$

(B) $h'(1) = 1216$

(C) $g'(1) = \frac{1}{5}$

(D) $h(g(2)) = 19$

6. There are 20 points lying on a circle, distance between two consecutive points is same. Three points are selected at random. X represent no. of ways to select them. Y represent no. of ways to select them such that no two adjacent objects are selected and Z represent no. of ways to select such that chosen objects are not diametrically opposite then

(A) $X = 1140$

(B) $Y = 800$

(C) $Z = 960$

(D) $Y \cap Z = 660$

7. An ellipse having eccentricity $\frac{3}{4}$ passes through the point P (5, 4). If the nearer focus is (1,1) and equation of tangent at P on the ellipse $4x + 3y - 32 = 0$, then

(A) Length of latus rectum of ellipse is 17.5

(B) Length of chord through (1,1) parallel to the tangent at P is 17.5

(C) Length of perpendicular from the focus other than (1, 1) on the tangent at P is 35

(D) Locus of point of intersection of perpendicular tangents drawn to ellipse is $(x + 1)^2 + (y + 8)^2 = 575$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
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Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let f(x) is strictly increasing and continuous function in [0, 1] and for $x \in [0, 1]$ $g(x) = \int_0^x f(t)\,dt$, also $\int_0^1 (f(t) + |f(t)|)\,dt = 6$. Local minimum value of g(x) is -1 occurs at x = a, where $a \in (0, 1)$, then

1. $\int_0^1 g(x) f(x)\,dx$ is equal to

2. $\int_a^1 f(x)\,dx$ is equal to

Paragraph for Questions 3 and 4

A tetrahedron is formed by the planes $P_1 : y + z = 0$. $P_2 : z + x = 0$, $P_3 : x + y = 0$ and $P_4 : x + y + z = 3$.

3. The volume V of the tetrahedron is :

4. Square of distance of P_4 from the vertex opposite to it is

SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
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Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

---

5. Consider $f(x) = \dfrac{x^2+4}{|x|}$ and $g(x) = p|x| + 2$, $p > 0$. If the exhaustive set of values of 'p' for which $f(x) > g(x)\ \forall\ x \in R - \{0\}$ is (a, b) then a + 4b is

6. A fair coin is tossed 10 times and the outcomes are listed. Let E_i be the event that the ith outcome is a head and E_m be the event that the list contains exactly m heads. If E_i and E_m are independent, then 'm' is equal to

7. A perpendicular is drawn from a point on the line $\dfrac{x-1}{2} = \dfrac{y+1}{-1} = \dfrac{z}{1}$ to the plane $x + y + z = 3$ such that the foot of the perpendicular Q also lies on the plane $x - y + z = 3$, if the coordinate of Q is (a, b, c) then $|a - b - c|$ is

8. If $\vec{a}_1 = a\hat{i} + b\hat{j}$, $\vec{a}_2 = \vec{a}_1 - \left(2\hat{i} + \sqrt{5}\hat{j}\right)$ and $\vec{a}_3 = \vec{a}_1 + \left(2\hat{i} - \sqrt{5}\hat{j}\right)$, then

$$\begin{vmatrix} 1+\vec{a}_1\cdot\vec{a}_1 & 1+\vec{a}_1\cdot\vec{a}_2 & 1+\vec{a}_1\cdot\vec{a}_3 \\ 1+\vec{a}_2\cdot\vec{a}_1 & 1+\vec{a}_2\cdot\vec{a}_2 & 1+\vec{a}_2\cdot\vec{a}_3 \\ 1+\vec{a}_3\cdot\vec{a}_1 & 1+\vec{a}_3\cdot\vec{a}_2 & 1+\vec{a}_3\cdot\vec{a}_3 \end{vmatrix}$$ is equal to

9. If $\sum_{r=0}^{3n} a_r(x-4)^r = \sum_{r=0}^{3n} A_r(x-5)^r$ and $a_k = 1\ \forall\ k \geqslant 2n$ and $\sum_{r=0}^{3n} d_r(x-8)^r = \sum_{r=0}^{3n} B_r(x-9)^r$ and $\sum_{r=0}^{3n} d_r(x-12)^r = \sum_{r=0}^{3n} D_r(x-13)^r$ and $d_k = 1\ \forall\ k \geqslant 2n$, then the value of $\dfrac{A_{2n} + D_{2n}}{B_{2n}}$.

10. If θ is the angle between $\bar{a}$ and $\bar{b}$ such that $\sqrt{3}(\vec{a}\cdot\vec{b}) + \left|\vec{b} - (\vec{a}\cdot\vec{b})\vec{a}\right| = 1 + \vec{b}\cdot\vec{b}$ where $|\bar{a}| = 1$ then $\text{cosec}^2\theta$ is

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work



ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Advanced)
FULL SYLLABUS
19-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	C	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,C	B,C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	A	B	C	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	2	5	3	2	60	2

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	D	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	C,D	A,B,C	A,B,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	B	D	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	4	12	85	20	1140	39

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	A	A	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C	A,C	B,C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	B	D	A	D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	1	6	9	31	8	23

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (D)

Acceleration $a = \frac{dv}{dt}$

$\Rightarrow a = 2bt$

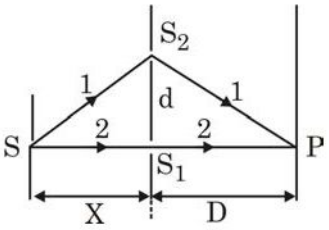
For block to just slide on the plate, we have

$\mu mg = m(2bt)$

$\Rightarrow t = \frac{\mu g}{2b}$

2. Ans (A)

Refer to figure, to reach point P, wave 1 has to travel a path ($SS_2 + S_2P$) while wave 2 has to travel a path ($SS_1 + S_1P$). Therefore, when the waves arrive at P, the path difference is



$\delta = (SS_2 + S_2P) - (SS_1 + S_2P)$

Now, in triangle SS_2S_1, we have .....(1)

$SS_2 = (x^2 + d^2)^{1/2} = \left(1 + \frac{d^2}{x^2}\right)^{1/2}$

$= \left(1 + \frac{d^2}{2x^2}\right)$ ($\because d << x$)

Similarly, $S_2P = (D^2 + d^2)^{1/2} = D\left(1 + \frac{d^2}{2D^2}\right)$

($\because d << D$)

Also $(SS_1 + S_1P) = x + D$, Using these in Eq. (1), we have

$\Delta = x\left(1 + \frac{d^2}{2x^2}\right) + D\left(1 + \frac{d^2}{2D^2}\right) - (x + D)$

$= x + \frac{d^2}{2x} + D + \frac{d^2}{2D} - x - D$

or $\Delta = \frac{d^2}{2}\left(\frac{1}{x} + \frac{1}{D}\right)$

In order to have a dark fringe at P, $\Delta = \frac{\lambda}{2}$. Hence

$\frac{\lambda}{2} = \frac{d^2}{2}\left(\frac{1}{x} + \frac{1}{D}\right)$ .....(1)

or $d = \left[\frac{\lambda x D}{x + D}\right]^{1/2}$ .....(2)

Putting $x = \frac{D}{2}$ in Eq. (2), we find that the correct option is (A).

4. Ans (A)

Maximum speed is attained when the entire energy stored in inductor is converted into kinetic energy of rod.

$\Rightarrow \frac{1}{2}L\left(\frac{E}{R}\right)^2 = \frac{1}{2}mv^2$

$\Rightarrow v = \left(\frac{E}{R}\right)\sqrt{\frac{L}{m}}$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C,D)

$\Rightarrow dI = kr^2 (2\pi r dr)$

$\Rightarrow I = 2\pi k \int_0^a r^3 dr$

$\Rightarrow I = \frac{1}{2}\pi k a^4$

Further, field for $r > a$ is

$B = \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 k a^4}{4r}$

and field for $r < a$ is

$B = \frac{\mu_0\left(\frac{\pi k r^4}{2}\right)}{2\pi r} = \frac{1}{4}\mu_0 k r^3$

6. Ans (A,C)

When number of electrons increases, then number of photons emitted also increases, thereby increasing the intensity. Energy, wavelength and frequency of photon are related to potential difference.

7. **Ans (B,C,D)**

Work done by $\vec{F}_1$ is $W_1 = \int_{P_1}^{P_2} F_1 \cos\theta ds$

Here, $ds = (6)d(-2\theta) = -12\, d\theta$ and $F_1 = 20$ N

$\therefore W_1 = -240 \int_{x/4}^{0} \cos\theta d\theta = 240 \sin\frac{\pi}{4} = 120\sqrt{2}$ J

$\vec{F}_1$ is conservative because it is always directed towards a fixed point P_2 .

$W_2 = F_2(OP_2) = (30)(6) = 180$ J and

$W_3 = \vec{F}_1 \int_0^{6(\pi/2)} F_3 ds = \int_0^{3\pi} 15 ds$

$= [15x]_0^{3\pi} = 45\pi$

PART-1 : PHYSICS

SECTION-I (iii)

11. **Ans (B)**

disc gives uniform E so force on dipole is zero.

Electric field inside a sheet varies linely.

PART-1 : PHYSICS

SECTION-II

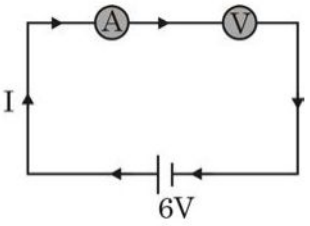
1. **Ans (2)**

Let R_1 = resistance of ammeter and R_2 = combined resistance of ammeter and voltmeter

In the first case,current inthe circuit is given by

$I = \frac{6}{R_2}$ .....(1)

and voltage across voltmeter is given by



V = 6 – (voltage across ammeter)

$\Rightarrow V = 6 - IR_1$

$\Rightarrow V = 6 - \frac{6R_1}{R_2}$ .....(2)

In the second case reading of ammeter becomes two times, i.e., the total resistance becomes half whereas the resistance of ammeter remains unchanged. Hence,

$I' = \frac{6}{R_2/2} = \frac{12}{R_2}$ .....(3)

and $V' = 6 - (I')R_1$

$\Rightarrow V' = 6 - \frac{12R_1}{R_2}$ .....(4)

Further, it is given that $V' = \frac{V}{2}$

$\Rightarrow 6 - \frac{12R_1}{R_2} = 3 - \frac{3R_1}{R_2}$

$\Rightarrow \frac{R_1}{R_2} = \frac{1}{3}$

Substituting this value in equation (4), we have

$V' = 6 - (12)\left(\frac{1}{3}\right)$

$\Rightarrow V' = 2V$

2. Ans (5)

For an adiabatic process $TV^{\gamma-1}$ = constant

$\Rightarrow TV^{\gamma-1} = \left(\frac{T}{2}\right)(5.66V)^{\gamma-1}$

$\Rightarrow (5.66\ V)^{\gamma-1} = 2$

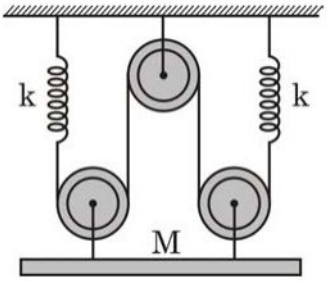
$\Rightarrow (\gamma - 1)\log_e(5.66) = \log_e(2)$

$\Rightarrow \gamma - 1 = 0.4$

$\Rightarrow \gamma = 1.4$

Since, $\gamma = 1 + \frac{2}{f} = 1.4$

4. Ans (2)



Net restoring force is

$F = -8x$

$\Rightarrow Ma = -8kx$

$\Rightarrow f = \frac{1}{2\pi}\sqrt{\left|\frac{a}{x}\right|}$

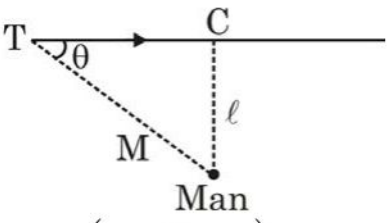
$\Rightarrow f = \frac{1}{2\pi}\sqrt{\frac{8k}{M}} = \frac{1}{\pi}\sqrt{\frac{2k}{M}}$

$\Rightarrow x = 2$

5. Ans (60)

Let the velocity of truck at T when it blows the whistle be v_s.

Then



$600 = \left(\frac{v}{v - v_s\cos\theta}\right)500$ .....(1)

During this time, speed of truck gets doubled, so

$2v_s = v_s + at$

$\Rightarrow v_s = at$ .....(2)

Now, $vt = AM = \frac{\ell}{\sin\theta}$ .....(3)

Also, $AB = \ell\cot\theta = v_s t + \frac{1}{2}at^2$ .....(4)

$\Rightarrow \frac{\ell\cos\theta}{\sin\theta} = at^2 + \frac{1}{2}at^2 = \frac{3}{2}at^2$

$\Rightarrow \frac{a\ell}{v} = \left(\frac{2}{3}\sin\theta\cos\theta\right)v$

From equation (1), we get

$\frac{6}{5} = \frac{u}{v - at\cos\theta} = \frac{v}{v - \frac{a\ell}{v}\left(\frac{\cos\theta}{\sin\theta}\right)}$

$\left\{\because\ t = \frac{\ell}{v\sin\theta}\right\}$

$\Rightarrow \frac{6}{5} = \frac{v}{v - \left(\frac{2}{3}\sin\theta\cos\theta\right)\left(\frac{\cos\theta}{\sin\theta}\right)v}$

$\Rightarrow \cos\theta = \frac{1}{2}$

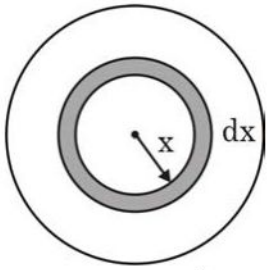
$\Rightarrow \theta = 60°$

6. Ans (2)

$$E = \frac{x}{2}\frac{dB}{dt}$$

$$\Rightarrow E = \frac{3Kxt^2}{2}$$

$$\Rightarrow d\tau = (dq)Ex = \left(\frac{3Kxt^2}{2}\right)\left(\frac{2\pi x dx}{\pi r^2}\right)qx$$



$$\Rightarrow \tau = \frac{3Kt^2q}{r^2}\int_0^r x^3dx$$

$$\Rightarrow \tau = \frac{3Kqt^2}{r}r^2 \text{ .....(1)}$$

torque due to friction force

$$d\tau = \mu g x dm = \mu g x(2\pi x dx)\frac{m}{\pi r^2}$$

$$\Rightarrow \tau = 2\mu g\frac{m}{r^2}\int_0^r x^2dx = \frac{2}{3}\mu mgr$$

$$\Rightarrow \frac{3Kqt^2r^2}{4} = \frac{2}{3}\mu mgr$$

$$\Rightarrow t = \sqrt{\frac{8\mu mg}{9Kqr}}$$

Substituting values, we get

$t = 2s$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (D)

On increasing volume concentration decreases.

4. Ans (B)

Q must be an ester. Addition of ethyl magnesium chloride to an alkyl acetate gives II, formate ester gives III and propionic ester gives IV. I can be produced from acetone and not from ester.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (C,D)

$$V = \frac{nRT}{P}$$

$V_A = 200R$, $V_C = 400R$, $V_B = 400R$

Obtaining equation for AB process

$$P = \frac{400R}{600R - V}$$

$$w_{AB} = -\int P\,dv$$

$= -\ 400R\ \ln 2$

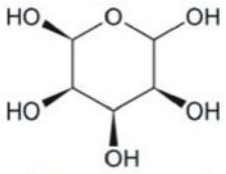
$$\Delta S = nC_V\ln\frac{T_2}{T_1} + nR\ln\frac{v_2}{v_1} = nCp\ln\frac{T_2}{T_1} + nR\ln\frac{P_1}{P_2}$$

$= 0.2$ L.atm

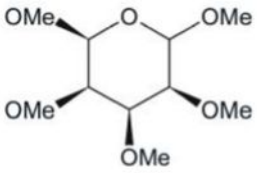
6. Ans (A,B,C)

Since anomeric carbon is a chiral carbon, so given sugar can exist in two anomeric pyranose forms.

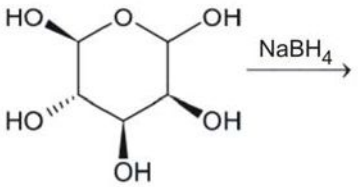
Since it is in hemiacetal, so it can reduce to tollen's reagent to silver mirror.

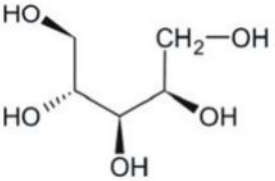


(Reducing sugar)



(Non-reducing because anomeric -OH is not - free





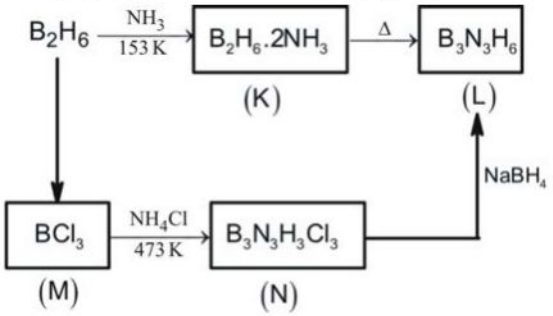
PART-2 : CHEMISTRY

SECTION-I (iii)

8. Ans (D)

K : $[BH_2(NH_3)_2]$ $[BH_4]$ M: BCl_3

L : $B_3N_3H_6$ N:$B_3N_3H_3Cl_3$

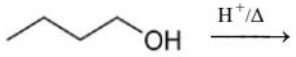
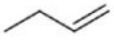


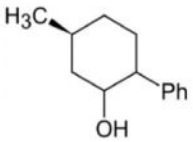
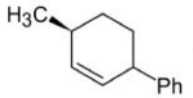
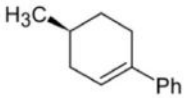
9. Ans (B)

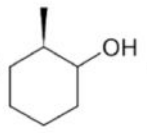
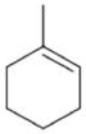
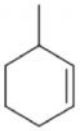
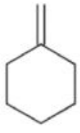
I^{st} order: $t_{1/2} = \frac{0.693}{k}$, $k = \frac{2.303}{t}$, $\log \frac{a}{a-x}$

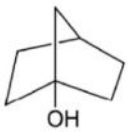
3 : $t_{1/2} = \frac{2}{k}\left(\sqrt{a} - \sqrt{\frac{a}{2}}\right)$

10. Ans (D)

(P)  (Major) +  (Minor)

(Q)   (Minor) +  (Major)

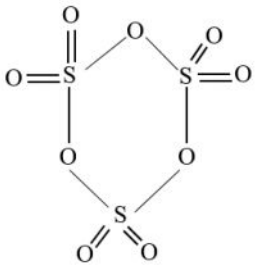
(R)   (Major) +  (Minor) +  (Minor)

(S)  cannot undergo dehydration as bridge head C-atom can never be sp^2-hybridised.

PART-2 : CHEMISTRY

SECTION-II

2. Ans (12)



x = 6, y = 6

3. Ans (85)

Let KI solution is M molar so m.mol = 20 × M

Given, KIO_3: $\frac{1}{10}$M

$\Rightarrow 20 \times M = \frac{1}{10} \times 30$

$\Rightarrow$ Molarity of KI = 0.15 M

$KIO_3 + 2KI + 6HCl \rightarrow 3IC\ell + 3kC\ell + 3H_2O$

m.mol: 5 10

$\Rightarrow$ AgI= 0.3 × 50−10=5 m.mol

$\Rightarrow$ % $AgNO_3$ = 85%

4. Ans (20)

$2[Au(CN)_2]^- + Zn \rightarrow 2Au + [Zn(CN)_4]^{2-}$

E° = −0.60−(−1.26) =0.66 V

Since E° value for the reduction of complexed Ag^+ ion to Ag is higher than that of complexed Au^+ ion into Au. So, Ag^+ ion will be reduced first.

Mole of Zn added = $\frac{78}{65} = 1.2$

Moles of $[Ag(CN)_2]^-$ = 0.003 × 500 = 1.5

Moles of $[Au(CN)_2]^-$ = 0.002 × 500 = 1.0

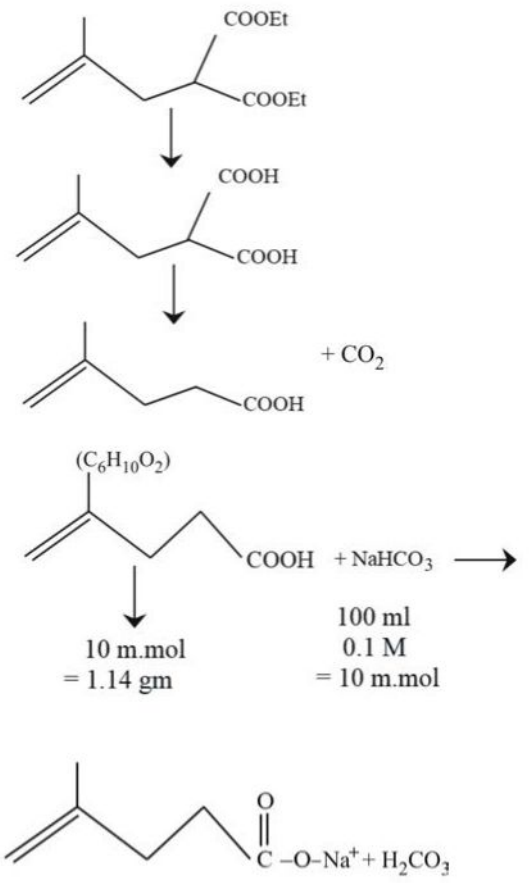
Moles of Zn used to reduce 1.5 moles of $[Ag(CN)_2]^-$ = 0.75

Moles of $[Au(CN)_2]^-$ reduced by remaining 0.45 moles of Zn = 0.90

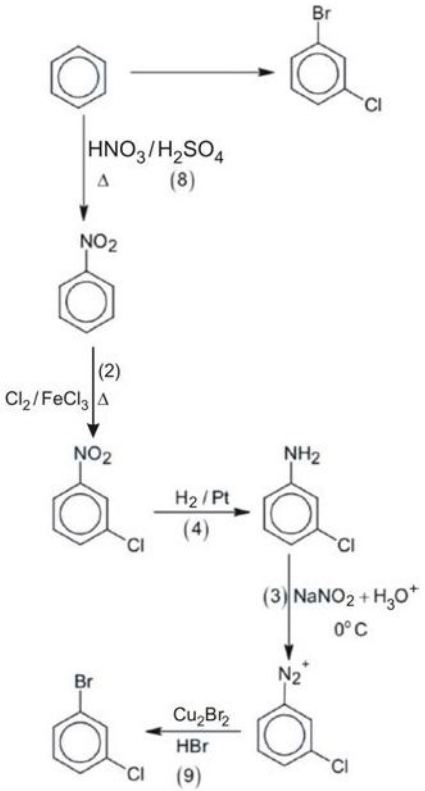
Moles of $[Au(CN)_2]^-$ left in the solution = 0.10

Concentration of $[Au(CN)_2]^-$ left in the solution = $\frac{0.10}{500} = 2 \times 10^{-4}M = x$

5. **Ans (1140)**



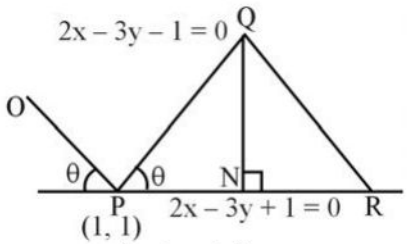
6. **Ans (39)**



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**



Let $Q = \left(t, \frac{2t-1}{3}\right)$

$PQ = QN \operatorname{cosec} \theta$

$\Rightarrow \quad t = \frac{47}{13}, -1$

For $t = -1$, Q lies on OP produced which is not possible

$\Rightarrow \quad Q \equiv \left(\frac{47}{13}, \frac{27}{13}\right)$

Let $R = \left(\frac{3\alpha - 1}{2}, \alpha\right)$

Slope of QR = slope of OP $\Rightarrow \alpha = 53/13$

2. **Ans (A)**

$S_1 < 0 \Rightarrow m^2 - 3m + 2 < 0 \Rightarrow m \in (1, 2)$

$(2m - 1 - 2)\left(2\left(\frac{3}{2}\right) - 0 - 2\right) < 0 \Rightarrow m < \frac{3}{2}$

hence, $m \in \left(1, \frac{3}{2}\right)$

3. **Ans (A)**

$h'(x) = \left(2x^2 - \ln x\right) g(x)$

$g'(x) = \frac{1}{\ln x^3} 3x^2 - \frac{1}{\ln x^2} 2x$

$g'(x) = \frac{x(x-1)}{\ln x} > 0 \forall x > 1; g(x) > g(1) \Rightarrow g(x) > 0 \forall x > 1$

For h(x) is increasing

$h'(x) > 0 \Rightarrow 2x^2 - \ln x > 0$ as $(g(x) > 0)$

Let $H(x) = 2x^2 - \ln x$

$H'(x) = 4x - \frac{1}{x} = \frac{4x^2 - 1}{x} > 0$ where $x > 1$

$H(x) > H(1) \Rightarrow H(x) > 2$

$\therefore h'(x) > 0 \ \forall \ x \in (1, \infty)$

$\therefore$ h(x) is increasing on $(1, \infty)$

4. Ans (B)

The given equation is $x^3 + y^3 = a^3$ .Differentiating both sides w.r.t. x we have

$$\frac{dy}{dx} = \frac{-x^2}{y^2}$$

$$\left(\frac{dy}{dx}\right)_{(x',y')} = -\left(\frac{x'}{y'}\right)^2 \quad .....(1)$$

Equation of the tangent at (x', y') is $y - y'$

$$= -\left(\frac{x'}{y'}\right)^2 (x - x')$$

$$(x')^2 x + (y')^2 y = (x')^3 + (y')^3$$

This passes through (x_1, y_1). Therefore

$$(x')^2 x_1 + (y')^2 y_1 = (x')^3 + (y')^3 = a^3 = x_1^3 + y_1^3$$

As (x_1, y_1) lies on the curve. Hence

$$x_1 (x'^2 - x_1^2) = -y_1 (y'^2 - y_1^2)$$

$$-\frac{x_1 (x' + x_1)}{y_1 (y' + y_1)} = \frac{y' - y_1}{x' - x_1}$$

= Slope of the line joining (x', y') and (x_1, y_1)

Which is the slope of the tangent at (x', y'). So from equation (i)

$$\frac{-x_1 (x' + x_1)}{y_1 (y' + y_1)} = -\left(\frac{x'}{y'}\right)^2$$

$$x_1 x' (y')^2 + x_1^2 (y')^2 = y_1 y' x'^2 + y_1^2 x'^2$$

$x' y'(x_1 y' - x' y_1) = -(x_1 y' - x' y_1)(x_1 y' + x' y_1)$ .... (ii)

Suppose $x_1 y' - x' y_1 = 0$ so that

$$\frac{x'}{x_1} = \frac{y'}{y_1} = \lambda$$

Hence, $x' = \lambda x_1$, $y' = \lambda y_1$ so that

$$a_3 = x'^3 + y'^3 = \lambda^3 (x_1^3 + y_1^3) = \lambda^3 a^3$$

Which implies $\lambda = 1$ and hence $x' = x_1$ and y_1

= y' a contradiction. Therefore $x_1 y' - x' y_1 \neq 0$ .Hence from equation (ii)

$$x' y' = -(x_1 y' + x' y_1)$$

Dividing both sides with x' y' we have

$$\frac{x_1}{x'} + \frac{y_1}{y'} = -1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C)

$f(x) = \sin x + px + q$

$f'(x) = \cos x + p$

If $p > 1$ then f(x) is increasing. So only one real root, which is positive if $q < 0$ and negative if $q > 0$.

If $p < -1$ f(x) is decreasing so only one real root, which is negative if $q < 0$

6. Ans (A,C)

$\vec{u}.\vec{w} = \cos\alpha \cos\gamma + \sin\alpha \sin\gamma = \cos(\alpha - \gamma)$

$\vec{v}.\vec{w} = \cos(\beta - \gamma)$, $\vec{u}.\vec{v} = \cos(\alpha - \beta)$

Let $A = \alpha - \gamma$, $B = \beta - \gamma$, $\alpha - \beta = A - B$

Now, $\vec{u}.\vec{w} = \vec{v}.\vec{w}$

$\Rightarrow \cos A = \cos B \Rightarrow B = 2m\pi + A$ where $m \in I$.

But $-2\pi < A, B < 2\pi$

$\therefore B = A, B = -A, B = 2\pi - A$ or $B = -2\pi + A$

when $B = A$ or $B = -2\pi + A$

we get $\cos A = -\sqrt{3}\cos 0$ or $-\sqrt{3}\cos(-2\pi)$

$\Rightarrow \cos A = -\sqrt{3}$. Not possible

when $B = -A$ or $B = 2\pi - A$, we get

$\cos A = -\sqrt{3}\cos(2A)$

$\Rightarrow 2\sqrt{3}\cos^2 A + \cos A - \sqrt{3} = 0$

$$\Rightarrow \cos A = \frac{-1 \pm \sqrt{1 + 4\left(2\sqrt{3}\right)\left(\sqrt{3}\right)}}{2\left(2\sqrt{3}\right)}$$

$$= \frac{-1 \pm 5}{4\sqrt{3}} = \frac{1}{\sqrt{3}}, \frac{\sqrt{3}}{2}$$

Now, $\vec{u}.\vec{v} = \cos(2A)$

$$= 2\cos^2 A - 1 = \begin{cases} -\frac{1}{3} \text{ when } \cos A = \frac{1}{\sqrt{3}} \\ \frac{1}{2} \text{ when } \cos A = -\frac{\sqrt{3}}{2} \end{cases}$$

7. Ans (B,C,D)

$3PQP^{-1} + 3P = 2P^{-1}QP + 2P$

$\Rightarrow 3PQP + 3PPP^{-1} = 2P^{-1}QP + 2P^{-1}PP$

$\Rightarrow |3P(Q+P)P^{-1}| = |2P^{-1}(Q+P)P|$

$\Rightarrow 3^n|P+Q| = 2^n|P+Q|$

$\Rightarrow |P+Q| = 0$

(c) $3PQP^{-1} - 3P^{-1}QP = -P^{-1}QP - P$

$= -P^{-1}QP - P^{-1}PP$

$\Rightarrow 3^n|PQP^{-1} - P^{-1}QP| = (-1)^n|Q+P| = 0$

PART-3 : MATHEMATICS

SECTION-I (iii)

8. Ans (B)

(P) $\frac{dy}{dx} = \frac{-y\sin x}{1+\cos x}$

$\Rightarrow \int \frac{dy}{y} = \int \frac{-\sin x}{1+\cos x}dx$

$\Rightarrow \ln y = \ln(1+\cos x) + \ln C$

$\Rightarrow y = (1+\cos x)C$

$\because y\left(\frac{\pi}{2}\right) = -1$

$\therefore y = -(1+\cos x) \Rightarrow y(0) = -2$

(Q) At $x = \frac{-\pi}{2}, -1, 0, 1, \frac{\pi}{2}$, f(x) is non-derivable

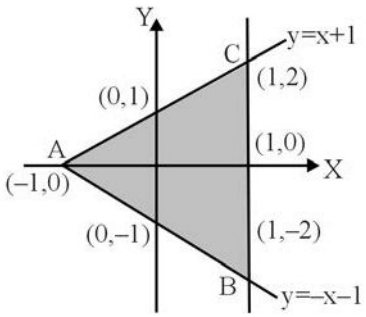
(R) Since $e^{\ln(x+1)} = x+1$, thus $e^{\ln(x+1)} = |y|$

$\Rightarrow x+1 \geq |y|$

$\Rightarrow x+1 < 0., x \geq 1$ or $(x+1) \geq 0$

and $-(x+1) \leq y \leq (x+1)$.

Thus the region represented by inequalities $-(x+1) < y < (x+1)$ and $|x| \leq 1., -1 \leq x \leq 1$ is as shown below



Thus required area is the area of ΔABC =

$\frac{1}{2}(4)(2) = 4$ square units

(S)

$\frac{(\sec^{-1}x + \text{cosec}^{-1}x)^3 - 3\sec^{-1}x\,\text{cosec}^{-1}x\,(\sec^{-1}x + \text{cosec}^{-1}x)}{\left(\frac{\pi}{2}\right)^3} = 7$

$\Rightarrow \frac{\pi^3}{8} - 3\sec^{-1}x\,\text{cosec}^{-1}x\left(\frac{\pi}{2}\right) = \frac{7\pi^3}{8}$

$\Rightarrow \sec^{-1}x = \pi$ or $\sec^{-1}x = \frac{-\pi}{2}$

$\Rightarrow x = -1$

9. Ans (D)

(P) $\sin A \sin 2A \sin 3A \sin 4A$

$= ax^2 + bx^3 + cx^4 + dx^5$

$a + b + c + d = f(1)$

Put $x = 1$

$\Rightarrow \sin^2 A = 1$

$A = (2n+1)\frac{\pi}{2}$

$\Rightarrow \sin A \cdot \sin 2A \cdot \sin 3A \cdot \sin 4A = 0$

(Q) $2 \sin x \cos x = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin 2x = \frac{1}{\sqrt{2}}$

$\Rightarrow 2x = \frac{\pi}{4}, \frac{3\pi}{4}$

$\Rightarrow x = \frac{\pi}{8}, \frac{3\pi}{8}$

$\Rightarrow d = \frac{3\pi}{8} - \frac{\pi}{8} = \frac{\pi}{4}$

(R) $4 \cos x \left(2 - 3\sin^2 x\right) + (\cos 2x + 1) = 0$

$\Rightarrow 4 \cos x \left(-1 + 3\cos^2 x\right) + 2\cos^2 x = 0$

$\Rightarrow 2 \cos x \left[-2 + 6\cos^2 x + \cos x\right] = 0$

$\Rightarrow 2 \cos x (2 \cos x - 1)(3 \cos x + 2) = 0$

$\Rightarrow \cos x = 0, \frac{1}{2}, \frac{-2}{3}$

In $x \in \left[0, \frac{\pi}{2}\right] \Rightarrow x = \frac{\pi}{2}, \frac{\pi}{3}$

Least deference between the roots

$= \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$

(S) $f(x) = \sqrt{\sin^4 x + 4\cos^2 x} - \sqrt{\cos^4 x + 4\sin^2 x}$

$= \sqrt{\sin^4 x + 4 - 4\sin^2 x} - \sqrt{\cos^4 x + 4 - 4\cos^2 x}$

$= 2 - \sin^2 x - \left(2 - \cos^2 x\right)$

$= \cos^2 x - \sin^2 x = \cos 2x$

$g(\sin 2t) = \sin t + \cos t$

$(g(\sin 2t))^2 = 1 + \sin 2t$

$\Rightarrow (g(\sin 2t))^2 = 1 + \sin 2t$

$\Rightarrow g(\sin 2t) = \sqrt{1 + \sin 2t}$

$\therefore g(x) = \sqrt{1+x}, -1 \leqslant x \leqslant 1$

Now, $g\{f(x)\} = g(\cos 2x)$

$= \sqrt{1 + \cos 2x} = \sqrt{2}|\cos x|$

$\therefore$ Range of $g\{f(x)\}$ is $[0, \sqrt{2}]$

$\Rightarrow a^2 + b^2 = 2$

10. Ans (A)

(P)

$$\lim_{h\to 0^+} 2\int_0^1 \frac{h\,dx}{h^2 + x^2} = \lim_{h\to 0^+} \left[2\frac{h}{h}\tan^{-1}\frac{x}{h}\right]_{0^+}^{1} = \pi$$

(Q) Put $x - 2 = x$, $y + 1 = y$

The required region is defined by $1 \leqslant |x| + |y| \leqslant 2$

Required area $= \left(2\sqrt{2}\right)^2 - \left(\sqrt{2}\right)^2 = 6$

$$\text{(R)} \int_0^{110\pi} [\sin x + \cos x]\,dx = 55\int_0^{2\pi} [\sin x + \cos x]\,dx$$

$$= 55\int_{\frac{\pi}{4}}^{\frac{9\pi}{4}} \left[\sqrt{2}\sin\theta\right] d\theta$$

$= -55\pi$

$$\text{(S)} \frac{\left(\frac{x-2y+3}{\sqrt{5}}\right)^2}{4} + \frac{\left(\frac{2x+y+1}{\sqrt{5}}\right)^2}{\frac{1}{4}} = 1$$

$a^2 = 4, b^2 = \frac{1}{4} \Rightarrow e = \frac{\sqrt{15}}{4}$

$k^2 = 4a^2e^2 = 4 \times 4 \times \frac{15}{16} = 15$

11. **Ans (D)**

(P) $9 - a^2 > 5$

$\Rightarrow -2 < a < 2 \Rightarrow$ number of integral values = 3

(Q) $b + c = a^3 - a$ & $bc = a^2$

$x^2 - (a^3 - a)\,a + a^2 = 0$ has two roots b & c

as b & c are real so

$D \geq 0$

$(a^3 - a)^2 - 4a^2 \geq 0$

$a^2[(a^2 - 1)^2 - 4] \geq 0$

$a^2[a^4 - 2a^2 - 3] \geq 0$

$a^2(a^2 + 1)(a^2 - 3) \geq 0$

$a^2 - 3 \geq 0$

$a^2 \geq 3 \Rightarrow$ Least value of a^2 is 3.

(R) number of terms in $(a + b + c - d)^n$ is ${}^{n+3}C_3$

and the sum of coefficients = 2^n

So $2^n > {}^{n+3}C_3$ which is true for $n > 6$

(S) $\lim\limits_{x\to\infty} \left(8x^3 + ax^2\right)^{\frac{1}{3}} - bx$

$= \lim\limits_{x\to\infty}(2x)\left(1 + \dfrac{a}{8x}\right)^{\frac{1}{3}} - bx$

$= \lim\limits_{x\to\infty} 2x\left(1 + \dfrac{a}{24x}\right) - bx$

$= \lim\limits_{x\to\infty}(2 - b)x + \dfrac{a}{12}$

for limit to exists $2 - b = 0$

$\Rightarrow b = 2$ & $\dfrac{a}{12} = 1 \Rightarrow a = 12$

$a - 3b = 6$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1)**

Put $x^2 = X \Rightarrow x\,dx = \dfrac{dX}{2}$

$y^2 = Y \Rightarrow y\,dy = \dfrac{dY}{2}$

$(2X + 3Y - 7)\dfrac{dX}{2} - (3X + 2Y - 8)\dfrac{dY}{2} = 0$

$\dfrac{dY}{dX} = \dfrac{2X + 3Y - 7}{3X + 2Y - 8}$

Reduce it to variable separable by substituting

$X = X' + h$

$Y = Y' + k$

We get the solution

$\left(x^2 + y^2 - 3\right) = \left(x^2 - y^2 - 1\right)^5 C$

2. **Ans (6)**

$\left||z| - \left|\dfrac{5}{z}\right|\right| \leqslant \left|z - \dfrac{5}{z}\right| \Rightarrow \left||z| - \dfrac{5}{|z|}\right| \leqslant 4$

Let $|z| = t$

$-4 \leq t - \dfrac{5}{t} \leqslant 4 \Rightarrow -4t \leqslant t^2 - 5 \leqslant 4t$

Now, $t^2 + 4t - 5 \geq 0$ $\quad t^2 - 4t - 5 \leq 0$

$(t + 5)(t - 1) \geq 0$ $\qquad (t - 5)(t + 1) \leq 0$

$\Rightarrow t \in [1,\infty)$ $\quad t \in [0, 5]$ (as t is positive)

$\therefore t \in [1,5]$

3. **Ans (9)**

$\left(\dfrac{x^4 + x^2 + 1}{x^2 - x + 1} - \dfrac{x^2 + 1}{x}\right)^{12} = \left(x^2 - \dfrac{1}{x}\right)^{12}$

$\Rightarrow 9^{th}$ term is free from 'x'

4. Ans (31)

The key is to count backwards. First choose the pair which you pick on Wednesday in 5 ways. Then there are four pairs of socks for you to pick a pair of on Tuesday and don't want to pick a pair. Since there are 4 pairs, the number of ways to do this is $\binom{8}{2}-4$. Then, there are two pairs and two nonmatching socks for you to pick from on Monday a total of 6 socks. Since you don't want to pick a pair, the number of ways to do this $\binom{6}{2}-2$. Thus the answer is

$$\frac{(5)\left(\binom{8}{2}-4\right)\left(\binom{6}{2}-2\right)}{\binom{10}{2}\binom{8}{2}\binom{6}{2}}=\frac{26}{315}$$

$$\frac{m+n}{11}=\frac{315+26}{11}=\frac{341}{11}=31$$

5. Ans (8)

$\because$ Elements of determinant can be 1 or -1

$\therefore$ First two places of first row can be filled in $2\times 2 = 4$ ways and last place can be filled in only one way to product be 1

$\therefore$ First row can be made in 4 ways

Similarly, second row can be made in 4 ways and third row can be made in 1 ways only

$\therefore\ m = 16$

$$\because\ |P| = \begin{vmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = 1$$

$\because$ $\text{adj}(\text{adj}P) = |P|^{n-2}.P = P \Rightarrow \text{tr}(\text{adj}(\text{adj } P)) = \text{tr}(P) = 2 = n$

$$\because\ \frac{m}{n}=\frac{16}{2}=8$$

6. Ans (23)

Let the position vectors of A, B and C be $\vec{O}, \vec{b}$ and $\vec{c}$-respectively.

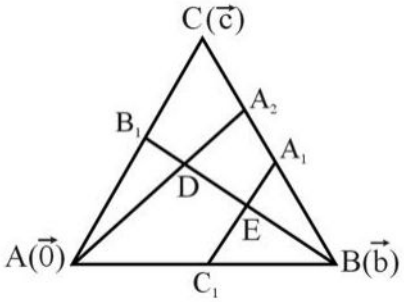
We have, $\overrightarrow{AC_1}=\frac{\vec{b}}{2}$, $\overrightarrow{AB_1}=\frac{\vec{c}}{2}$, $\overrightarrow{AA_1}=\frac{\vec{b}+\vec{c}}{2}$, $\overrightarrow{AA_1}=\frac{3\vec{c}+\vec{b}}{4}$

Equation of lines BB_1, AA_2 and C_1A_1 are respectively

$$\vec{r}=\vec{b}+\lambda_1\left(\frac{\vec{c}}{2}-\vec{b}\right)$$

$$\vec{r}=\lambda_2\frac{3\vec{c}+\vec{b}}{4} \text{ and } \vec{r}=\frac{\vec{b}}{2}+\lambda_3\left(\frac{\vec{c}}{2}\right)$$

For point D, we have $\vec{b}+\lambda_1\left(\frac{\vec{c}}{2}-\vec{b}\right)=\lambda_2\left(\frac{3\vec{c}+\vec{b}}{4}\right)$



$$\Rightarrow\ \vec{b}\left(1-\lambda_1-\frac{\lambda_2}{4}\right)+\frac{\vec{c}}{4}(2\lambda_1-3\lambda_2)=\vec{0}$$

$$\Rightarrow\ \lambda_1=\frac{6}{7},\lambda_2=\frac{4}{7}$$

$$\Rightarrow\ \overrightarrow{AD}=\frac{3\vec{c}+\vec{b}}{7}$$

For point E we have $\vec{b}+\lambda_1\left(\frac{\vec{c}}{2}-\vec{b}\right)=\frac{\vec{b}}{2}+\frac{\lambda_3}{2}\vec{c}$

$$\Rightarrow\ \lambda_1=\lambda_3=\frac{1}{2}\Rightarrow=\frac{2\vec{b}+\vec{c}}{4}$$

Now, $\overrightarrow{EA}_1=\frac{3\vec{c}+\vec{b}-2\vec{b}-\vec{b}}{4}=\frac{2\vec{c}-\vec{b}}{4}$,

$$\overrightarrow{DA}_1=\frac{\vec{b}+\vec{c}}{2}-\frac{3\vec{c}+\vec{b}}{7}=\frac{5\vec{b}+\vec{c}}{14}$$

Area of the quadrilateral $EA_1A_2D=\frac{1}{2}\left|\overrightarrow{EA}_2\times\overrightarrow{DA}_1\right|$

$$=\frac{1}{112}\left|\left(2\vec{c}-\vec{b}\right)\times\left(5\vec{b}+\vec{c}\right)\right|=\frac{1}{112}\left|10\vec{c}\times\vec{b}-\vec{b}\times\vec{c}\right|$$

$$=\frac{11}{112}\left|\vec{c}\times\vec{b}\right|=\frac{11}{56}\times\frac{1}{2}\left|\vec{c}\times\vec{b}\right|=\frac{11}{56}\ \text{(area of }\Delta ABC)$$

Thus required ratio of is $\frac{11}{56}$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Advanced)
FULL SYLLABUS
19-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	A	C	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	A,C,D	A,B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	0.25	0.00	1.41	6.70 to 6.85		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	6	2	5	6	2	6

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	D	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C	A,C,D	B,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	8.00	6.00	130.00	365.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	5625	9	4	3	3	3

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	A	D		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	A,B,C,D	A,B,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.00	3.00	18.00	3.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	3	5	1	80	2	4

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

According to Newton's second Law, we have

$$F = \frac{\Delta p}{\Delta t} = \left(\frac{\Delta m}{\Delta t}\right)(\Delta v)$$

$$\Rightarrow F = \rho\left(\frac{\Delta V}{\Delta t}\right)(2v)$$

$$\Rightarrow F = \rho(Av)(2v) = 2\rho Av^2$$

2. Ans (A)

$$A = A_0 e^{-\lambda t} \Rightarrow \frac{6}{100} = e^{-\frac{\ell n 2}{T_{\frac{1}{2}}}t}$$

$$\frac{1}{\frac{t}{T_{\frac{1}{2}}}} = \frac{6}{100} \Rightarrow \frac{1}{2^4}$$

$$T_{\frac{1}{2}} = 30\,\text{min}$$

3. Ans (C)

$$\frac{dA}{dt} = \frac{L}{2m}$$

4. Ans (B)

B_W is magnetic field of wire

$B_1 = B_0 + B_W$

$B_2 = B_0 - B_w$

$\Rightarrow B_0 = \frac{B_1 + B_2}{2}$

PART-1 : PHYSICS

SECTION-I (ii)

6. Ans (A,C,D)

(A) $U = \frac{1}{2}CV^2$

As source between the plates is connected, so potential difference remains constant. But capacitance C becomes KC hence energy stored is increased by factor K.

(B) electric field $\frac{V}{d}$ is not changed.

(C) Charge on each plate is increased by factor K hence force between them increases by a factor of K^2.

(D) Since Q = CV

Hence charge becomes KQ as C becomes KC and V remains unchanged.

7. Ans (A,B,C,D)

Let ℓ_1 and ℓ_2 be the length of the air column for the first and the second resonance respectively with a tuning fork of frequency f.

Then $\lambda = 2(\ell_2 - \ell_1)$

From relation, $v = f\lambda$

$v = 2f(\ell_2 - \ell_1)$

$= 2 \times 480 \times (51\text{ cm} - 16\text{ cm}) = 336$ m/s

If T be the temperature of the air in air column of the resonance tube.

Then from formula,

$$\frac{v}{v_a} = \sqrt{\frac{T}{T_0}}$$

$$T = T_0\left(\frac{v}{v_0}\right)^2 = 273\left(\frac{336}{330}\right)^2 = 283K$$

Also $\frac{\lambda}{4} = \ell_1 + e = 16\text{ cm} + 0.3\text{ d}$

$$= \frac{v}{4f} = \frac{336}{4 \times 480}m$$

Therefore d = 5cm

Again

$$\frac{\Delta v}{v} \times 100\% = \frac{\Delta\ell_2 + \Delta\ell_1}{\ell_2 - \ell_1} \times 100\% = \frac{4}{7}\%$$

And

$$\frac{\Delta T}{T} \times 100\% = 2 \times \frac{\Delta\ell_2 + \Delta\ell_1}{\ell_2 - \ell_1} \times 100\% = \frac{8}{7}\%$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (0.25)

Velocity of particle undergoing wave motion is given by

$$\frac{\partial y}{\partial t} = -u\frac{\partial y}{\partial x}$$

At t = 0 and x = 2, we have

$$\frac{\partial y}{\partial t} = \frac{-1 \times 10 \times 10^{-3}}{4 \times 10^{-2}} = -\frac{1}{4} = -0.25\text{ ms}^{-1}$$

2. Ans (0.00)

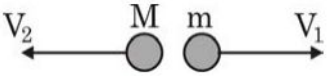
Displacement and velocity of particle is vector sum of displacements and velocity due to individual pulses.

So, displacement at x = 8 cm and t = 0.06 sec is

y = 5 mm – 5 mm = 0

3. Ans (1.41)

AT an instant when m is just leaving M, let v_1 and v_2 be the velocities of m and M (both horizontal).



By law of conservation of linear momentum, we have

$1 \times v_1 = 4v_2$ .....(1)

By law of conservation of Mechanical energy, we have

$(1)(10)(4-2) = \frac{1}{2}(1)v_1^2 + \frac{1}{2}(4)v_2^2$

$\Rightarrow v_1^2 + 4v_2^2 = 40$ .....(2)

Solving these two equations, we get

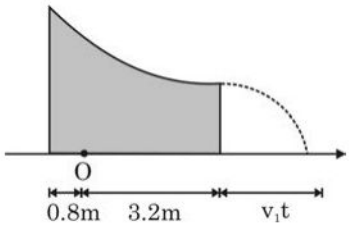
$v_2 = \sqrt{2}ms^{-1}$ and $v_1 = 4\sqrt{2}\ ms^{-1}$

4. Ans (6.70 to 6.85)

$\sum m_g x_L = \sum m_L x_L$

$\Rightarrow 4d = 1(4-d)$

$\Rightarrow d = 0.8$ m



i.e., m is already at x = 3.2 m at this instant. Time taken by m to fall to ground from this instant will be

$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 2}{10}} = \sqrt{\frac{2}{5}}s$

The x-coordinate of m where it strikes the ground will be

$x = 3.2 + v_1 t = 3.2 + 4\sqrt{2} \times \sqrt{\frac{2}{5}} = 6.8$ m

PART-1 : PHYSICS

SECTION-II (ii)

5. Ans (6)

$Q - \text{factor} = \frac{\omega_0}{2\Delta\omega}$

6. Ans (2)

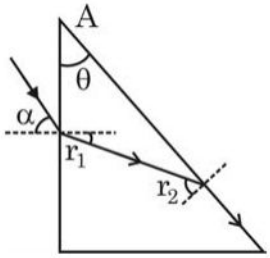
$\Delta P = \sigma\left(\frac{1}{D/2} + \frac{1}{\infty}\right) = \frac{2\sigma}{D}$

Net force = $\sigma\pi D - (\Delta P)$ area

$= \sigma\pi D - \frac{2\sigma}{D}\frac{\pi D^2}{4} = \frac{\sigma\pi D}{2}$

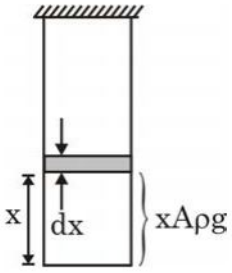
7. Ans (5)

Angle of prism, $A = r_2 - r_1$



8. Ans (6)

Consider an element as shown in the figure.



Stress in the element = $\frac{\text{Force}}{\text{Area}} = \frac{xA\rho g}{A} = x\rho g$

Now, elastic potential energy stored in the wire is

$dU = \frac{1}{2}$ (Stress) (Strain) (Volume)

$\Rightarrow dU = \frac{1}{2}\frac{(\text{Stress})^2}{Y}(\text{Volume})$

$\Rightarrow dU = \frac{1}{2}\frac{(x\rho g)^2}{Y}Adx = \frac{1}{2}\frac{\rho^2 g^2 A}{Y}x^2 dx$

Total elastic potential energy is

$U = \frac{1}{2}\frac{\rho^2 g^2 A}{Y}\int_0^L x^2 dx$

$U = \frac{\rho^2 g^2 A L^3}{6Y}$

$\Rightarrow p = 6$

10. Ans (6)

For equilibrium of cylinder in horizontal direction, we have

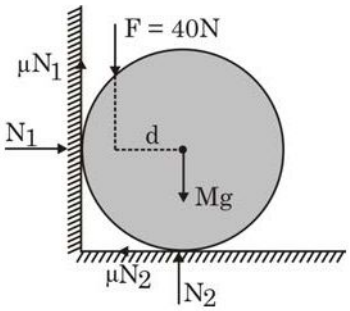
$N_1 = \mu N_2$ .....(1)

For equilibrium of cylinder in vertical direction, we have

$N_2 + \mu N_1 = F + Mg$ .....(2)

Solving these two equations with F = 40 N, M = 2kg and $\mu = \frac{1}{3}$, we get

$N_1 = 18$ N and $N_2 = 54$ N



Since, $Fd = \mu(N_1 + N_2)r$

$$\Rightarrow d = \frac{\mu(N_1 + N_2)r}{F} = \frac{\left(\frac{1}{3}\right)(18 + 54)(0.1)}{40}$$

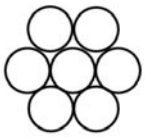
$\Rightarrow d = 0.06$ m = 6 cm

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)

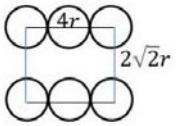
Plane P has the following arrangement of circular atoms:



Considering the smallest possible triangle in this arrangement

$$PF = \frac{\frac{1}{6} \times 3 \times \frac{22}{7}r^2}{\frac{1}{2} \times 2r \times \sqrt{3}r} = 0.907$$

And Plane Q has the arrangement:



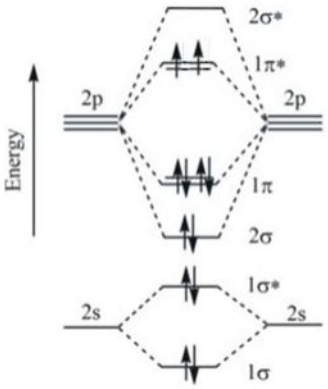
$$PF = \frac{\left[\frac{1}{2} \times 2 + \frac{1}{4} \times 4\right] \times \frac{22}{7}r^2}{4r \times 2\sqrt{2}r} = 0.5555$$

3. Ans (D)

Friedel-Craft acylation occur in the ring without deactivating –Cl group

4. Ans (C)

O_2 (16 e^-)



Naming of molecular orbitals are as per preference of formation of σ & π bonds respectively.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A,B,C)

(A) $z > 1 \frac{d_i}{d_r} > 1 \Rightarrow d_r < d_i$

(B) Always repulsions z > 1

(C) First law of thermodynamics is valid for real gas (or) ideal gas

(D) 'z' linearly varies with p.

6. Ans (A,C,D)

$3Cu + 8HNO_3 \rightarrow 2NO + 3Cu(NO_3)_2 + 4H_2O$

NO⇒ has 11 valence e^- with an unpaired

e^- in π *2p orbital

⇒ is diamagnetic in solid state due to dimerisation

⇒ it is neutral oxide

PART-2 : CHEMISTRY

SECTION-II (i)

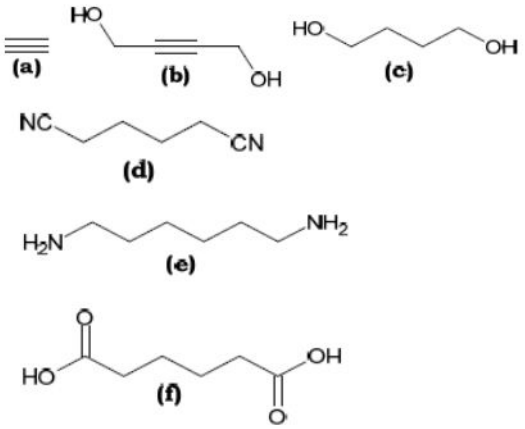
1. Ans (8.00)

$\Delta G = [2 \times (-14) + (-4)] - [2 \times (-16) + 8] = -8$ KJ/mol.

2. Ans (6.00)

$\Delta G = 3 \times (-14) - 3 \times (-16) = +6$ KJ/mol.

3. **Ans (130.00)**



Number of moles of hexamethylene diamine (e)

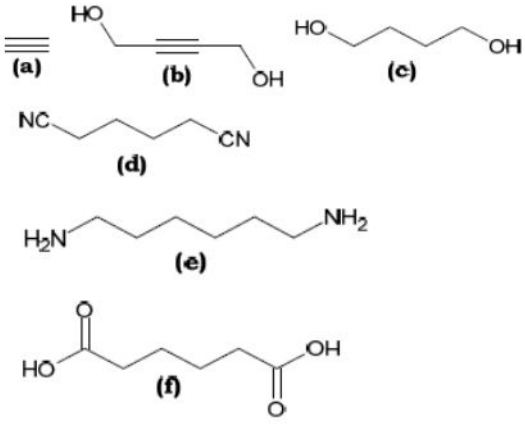
$= \frac{290}{116} = 2.5$

Number of moles of adipic acid (f) required = 2.5

Mass of adipic acid = 2.5 ×146 × 365

Number of moles of acetylene (a) required = 5

4. **Ans (365.00)**



Number of moles of hexamethylene diamine (e)

$= \frac{290}{116} = 2.5$

Number of moles of adipic acid (f) required = 2.5

Mass of adipic acid = 2.5 ×146 × 365

Number of moles of acetylene (a) required = 5

PART-2 : CHEMISTRY

SECTION-II (ii)

5. **Ans (5625)**

$\frac{\ell}{A} = \frac{2}{4}$

$R = \rho \cdot \frac{\ell}{A} \Rightarrow 100 = \rho \cdot \frac{1}{2}$

$k = \frac{1}{200}\Omega^{-1}\ cm^{-1}$

$\Lambda = \frac{1}{200} \times \frac{1000}{0.1} = 50$

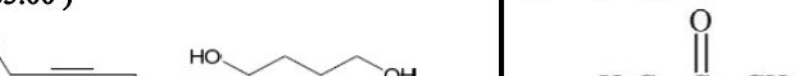
$\Lambda^\circ = 300 + 100 = 400$

$\alpha = \frac{\Lambda}{\Lambda^\circ} = \frac{50}{400} = \frac{1}{8}$

$\Delta T_b = i \times K_b \times m$

$= \left(1 + \frac{1}{8}\right) \times 0.5 \times 0.1 = 0.05625$

6. **Ans (9)**



$+ \ 3I_2 \ + \ 4NaOH \longrightarrow$

$CH_3COONa + 3CHI_3 + 3NaI$

7. **Ans (4)**

$[Fe(CN)_6]^{4-}$, $[Co(H_2O)_6]^{3+}$, $[PtCl_6]^{2-}$, $[Cr(CO)_6]$

8. **Ans (3)**

Correct : 1, 3 & 7

$\Rightarrow$ number of correct statements = 3

Incorrect : remaining 4

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

Put $x = 0$, $f(0)\cdot e^{f(0)} = 0$

$\Rightarrow \quad f(0) = 0$

$\Rightarrow$ f is bijective so $f(x) > 0 \ \forall \ x \in (0, \infty)$

$f(x)\, e^{f(x)} = x$

In $f(x)+f(x) = \ln x$

$$\frac{f(x)}{\ln x} = 1 - \frac{\ln f(x)}{\ln x} \quad \ldots (1)$$

We know that $e^x > x$

$x \longrightarrow f(x)$

$e^{f(x)} > f(x)$

$f(x)\, e^{f(x)} > f(x)\cdot f(x)$

$x > f^2(x)$

$\sqrt{x} > f(x)$

$$\frac{\sqrt{x}}{x} > \frac{f(x)}{x} \Rightarrow 0 < \frac{f(x)}{x} < \frac{1}{\sqrt{x}}$$

$$\lim_{x\to\infty} \frac{f(x)}{x} = 0$$

$\Rightarrow$ growth of f < growth of x

$\Rightarrow$ growth of ln f(x) < growth of ln x

$$\Rightarrow \lim_{x\to\infty} \frac{\ln f(x)}{\ln x} = 0$$

Putting this value in equation (1)

$$\lim_{x\to\infty} \frac{f(x)}{\ln x} = 1$$

2. Ans (C)

$x + xy + xyz = 1 \Rightarrow x(1 + y) = 1 - xyz \quad \ldots(1)$

$y + yz + xyz = 2 \Rightarrow y(1 + z) = 2 - xyz \ \ldots(2)$

$z + zx + xyz = 4 \Rightarrow z(1 + x) = 4 - xyz \ \ldots(3)$

Multiply all above equation

$xyz\ (\ 1 + x)(1 + y)(1 + z) = (1 - xyz)(2 - xyz)$

$(4 - xyz)$

Let

$xyz = p, \ (1 + x)(1 + y)(1 + z) = q$

$pq = (1 - p)(2 - p)(4 - p)$

Now $q = 1 + x + y + z + xy + yz + zx + xyz$

$q = 1 + x(1 + y) + y(1 + z) + z(1 + x) + p$

$q = 1 + 1 - p + 2 - p + 4 - p + p$

$q = 8 - 2p$

$p(8 - 2p) = (1 - p)\ (2 - p)\ (4 - p)$

$(4 - p)(2p - (1 - p)(2 - p)) = 0$

$(4 - p)\ (p^2 - 5p + 2) = 0$

$$p = 4, \ \frac{5 + \sqrt{17}}{2} \cdot \frac{5 - \sqrt{17}}{2}$$

Largest value of $xyz = \dfrac{5 + \sqrt{17}}{2}$

$a = 5, \ b = 17, \ c = 2$

$a + b + c = 24$

3. Ans (A)

$y(x + y)^2 = x$

d. w. r. to x

$2y\ (x + y)\ (1 + y') + (x + y)^2\ y' = 1$

$\Rightarrow y'\ (2y\ (x + y) + (x + y)^2) = 1 - 2y\ (x + y)$

$\Rightarrow (x + y)\ y'\ (2y + x + y) = 1 - 2y\ (x + y)$

$\Rightarrow (x + y)\ y'(x + 3y) + (x + y)\ (\ x + 3y)$

$= 1 - 2y\ (x + y) + (x + y)(x + 3y)$

$\Rightarrow (x + 3y)(x + y)\ (y' + 1)$

$= 1 + (x + y)\ (x + 3y - 2y)$

$\Rightarrow (x + 3y)\ ((x + y)y' + 1) = 1 + (x + y)^2$

$$\int \frac{1}{x + 3y}\,dx = \int \frac{(x + y)\,(y' + 1)}{1 + (x + y)^2}\,dx$$

$$\int \frac{1}{x + 3y}\,dy = \frac{1}{2}\ln\left((x + y)^2 + 1\right) + C$$

4. Ans (D)

From given condition odd elements of the domain can associate with unequal odd elements of the co-domain and even elements of the domain can associate with unequal even number of the co-domain.

Number of ways if x is odd

$= 4!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right) = 9$

Number of ways if x is even

$= 4!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right) = 9$

Total number of such functions = 9 x 9 = 81

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$f(x) = x^3 + 5x + 1$

$g'(y) = \frac{1}{f'(x)} = \frac{1}{3x^2+5}$

when y = 1 then x = 0

$\therefore \quad g'(1) = \frac{1}{5}$

$h(g(g(x)) = x$

$\therefore \quad x \longrightarrow f(x)$

$h(g(x)) = f(x)$

$\therefore \quad h(x) = f(f(x))$

$h(0) = f(f(0)) = 7$

$h(g(2))$

$f(f(g(2)) = f(2) = 8 + 10 + 1 = 19$

$h'(x) = f'(f(x)) \cdot f'(x)$

$h'(1) = f'(f(1)) \cdot f'(1)$

$= f'(7) \cdot f'(1)$

$= (147 + 5)(3 + 5) = 152 \times 8 = 1216$

6. Ans (A,B,C,D)

$X = {}^{20}C_3 = 1140$

$Y = \frac{{}^{20}C_1 \cdot {}^{16}C_2}{3} = 800$

$Z = {}^{20}C_3 - 10 \times {}^{18}C_1 = 1140 - 180 = 960$

$Y \cap Z = Y - 10 \times 14$

$= 800 - 140$

$= 660$

7. Ans (A,B,C)

$\because a(1 - e) = 5 \Rightarrow a = 20$

$\therefore b^2 = 400\left(1-\frac{9}{16}\right) = 175$

$\because P_1P_2 = b^2 \quad \therefore P_1 \times 5 = 175 \Rightarrow P_1 = 35$

also centre of elipse is $(-11, -8)$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (2.00)

Since g(x) has minimum value –1 at x = a

therefore $g(a) = \int_0^a f(t)\,dt = -1$ and

$g'(x) = f(a) = 0$.....(1)

Now since f(x) is increasing function in [0, 1]

$\Rightarrow f(x) < 0$ for $x \in [0, a)$ and $f(x) > 0$ for $x \in (a, 1]$

$\Rightarrow f(x) + |f(x)| = \begin{cases} 0 & 0 \leqslant x < a \\ 2f(x) & a \leqslant x \leqslant 1 \end{cases}$

$\Rightarrow \int_0^1 f(t) + |f(t)|dt = \int_0^a 0.dt + \int_0^1 2f(t)\,dt = 6$ (given)

$\Rightarrow \int_a^1 f(t)\,dt = 3$

2. Ans (3.00)

Since g(x) has minimum value –1 at x = a

therefore $g(a) = \int_0^a f(t)\,dt = -1$ and

$g'(x) = f(a) = 0$.....(1)

Now since f(x) is increasing function in [0, 1]

$\Rightarrow f(x) < 0$ for $x \in [0, a)$ and $f(x) > 0$ for $x \in (a, 1]$

$$\Rightarrow f(x) + |f(x)| = \begin{cases} 0 & 0 \leqslant x < a \\ 2f(x) & a \leqslant x \leqslant 1 \end{cases}$$

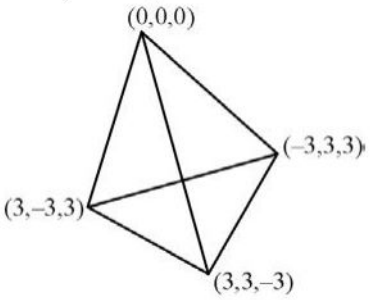
Also $g(1) = \int_0^1 f(t)\,dt$

$$= \int_0^a f(t)\,dt + \int_a^1 f(t)\,dt = -1 + 3 = 2$$

Now $\int_0^1 g(x) f(x)\,dx = \int_0^1 g(x) g'(x)\,dx = \left[\frac{g^2(x)}{2}\right]_0^1$

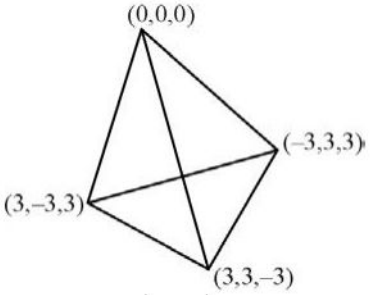
$$= \frac{g^2(1) - g^2(0)}{2} = \frac{2^2 - 0}{2} = 2$$

3. Ans (18.00)



$$V = \left|\frac{1}{6}\begin{vmatrix} 3 & -3 & 3 \\ -3 & 3 & 3 \\ 3 & 3 & -3 \end{vmatrix}\right| = \left|\frac{9}{2}\begin{vmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & -1 \end{vmatrix}\right| = 18$$

4. Ans (3.00)



Distance $= \left|\frac{3}{\sqrt{3}}\right| = \sqrt{3}$

PART-3 : MATHEMATICS

SECTION-II (ii)

5. Ans (3)

$f(x) > g(x)$

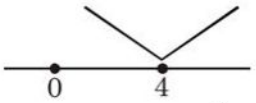
$$|x| + \frac{4}{|x|} > p|x| + 2 \quad \forall \; x \in R - \{0\}$$

$$p < \frac{|x| + \frac{4}{|x|} - 2}{|x|}$$

$$p < 1 + \frac{4}{(x)^2} - \frac{2}{|x|} \; \forall \; x \in R - \{0\}$$

Consider $h(x) = 1 + \frac{4}{x^2} - \frac{2}{x}$, $x > 0$

$$h'(x) = \frac{2(x-4)}{x^3}$$



Now, $h(4) = 1 + \frac{1}{4} - \frac{1}{2} = \frac{3}{4}$

$p < h(x)$

$\therefore \quad 0 < p < \frac{3}{4}$

$a = 0, \; b = 3/4$

$\therefore \quad a + 4b = 3$

6. Ans (5)

$P(E_i) = \frac{1}{2}$, $P(E_m) = {}^{10}C_m \left(\frac{1}{2}\right)^{10}$

$P(E_i \cap E_m) = \frac{1}{2}, {}^{9}C_{m-1}\left(\frac{1}{2}\right)^9 = \frac{{}^{9}C_{m-1}}{2^{10}}$

$\because$ E_i and E_m are independent

$\Rightarrow P(E_i \cap E_m) = P(E_i).P(E_m)$

$$\Rightarrow \frac{{}^{9}C_{m-1}}{2^{10}} = {}^{10}C_m \left(\frac{1}{2}\right)^{10} \cdot \frac{1}{2}$$

$\Rightarrow m = 5$

7. Ans (1)

Let point P on the line is $(2\lambda + 1, -\lambda - 1, \lambda)$

The foot of perpendicular Q is given by

$$\frac{x-2\lambda-1}{1} = \frac{y+\lambda+1}{1} = \frac{z-\lambda}{1} = \frac{-(2\lambda-3)}{3}$$

$\therefore$ Q Lies on $x + y + z = 3$ and $x - y + z = 3$

$\Rightarrow \quad x + z = 3$ and $y = 0$

$y = 0 \Rightarrow \lambda + 1 = \frac{-2\lambda+3}{3} \Rightarrow \lambda = 0$

$\Rightarrow$ Q is $(2, 0, 1)$

8. Ans (80)

$\vec{a}_1 = (a, b), \vec{a}_2 = (a-2, b-\sqrt{5}), \vec{a}_3 = (a+2, b-\sqrt{5})$

Area of Δ, whose vertices are $\vec{a}_1$, $\vec{a}_2$ and $\vec{a}_3$ is

$$\Delta = \frac{1}{2}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}$$

$$\Rightarrow \quad \Delta^2 = \frac{1}{4}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}$$

and

$|\vec{a}_1 - \vec{a}_2| = 3, |\vec{a}_1 - \vec{a}_3| = 3, |\vec{a}_2 - \vec{a}_3| = 4$

$\Rightarrow \Delta = \sqrt{5 \times 2 \times 2 \times 1} = 2\sqrt{5}$

$$(2\sqrt{5})^2 = \frac{1}{4}\begin{vmatrix} 1+\vec{a}_1\cdot\vec{a}_1 & 1+\vec{a}_1\cdot\vec{a}_2 & 1+\vec{a}_1\cdot\vec{a}_3 \\ 1+\vec{a}_2\cdot\vec{a}_1 & 1+\vec{a}_2\cdot\vec{a}_2 & 1+\vec{a}_2\cdot\vec{a}_3 \\ 1+\vec{a}_3\cdot\vec{a}_1 & 1+\vec{a}_3\cdot\vec{a}_2 & 1+\vec{a}_3\cdot\vec{a}_3 \end{vmatrix}$$

$\therefore$ Required answer $= 4 \times 5 \times 4 = 80$

9. Ans (2)

$x - 4 = t + 1 \Rightarrow x - 5 = t$

$\therefore \sum_{r=0}^{3n} a_r(1+t)^r = \sum_{r=0}^{3n} A_r t^r$

$a_0(1+t)^0 + a_1(1+t)^1 + \ldots + a_{2n}(1+t)^{2n} +$

$a_{2n+1}(1+t)^{2n+1} + \ldots + a_{3n}(1+t)^{3n}$

$= A_0t^0 + A_1t^1 + \ldots\ldots + A_{2n}t^{2n} + \ldots\ldots + A_{3n}t^{3n}$

compare coefficient of t^{2n} on both sides

$A_{2n} = a_{2n}\cdot {}^{2n}C_{2n} + a_{2n+1}\,{}^{2n+1}C_{2n} + \ldots + a_{3n}\,{}^{3n}C_{2n}$

$= {}^{2n}C_{2n} + {}^{2n+1}C_{2n} + {}^{2n+2}C_{2n} + \ldots + {}^{3n}C_{2n}$

$[\because \; a_{2n} = a_{2n+1} = \ldots = a_{3n} = 1]$

$= \left({}^{2n+1}C_{2n+1} + {}^{2n+1}C_{2n}\right) + \ldots + {}^{3n}C_{2n}$

$A_{2n} = {}^{3n+1}C_{2n+1}$

Similarly $B_{2n} = {}^{3n+1}C_{2n+1}$

$D_{2n} = {}^{3n+1}C_{2n+1}$

$\therefore \frac{A_{2n} + D_{2n}}{B_{2n}} = 2$

10. Ans (4)

$\sqrt{3}(\bar{a}\cdot\bar{b}) + |\bar{b} - (\bar{a}\cdot\bar{b})\bar{a}| = 1 + |\bar{b}|^2$

$\sqrt{3}(\bar{a}\cdot\bar{b}) + |\bar{a}\times(\bar{b}\times\bar{a})| = 1 + |\bar{b}|^2$

$\sqrt{3}(\bar{a}\cdot\bar{b}) + |\bar{a}|\,|\bar{b}\times\bar{a}|\sin 90^\circ = 1 + |\bar{b}|^2$

$\sqrt{3}(\bar{a}\cdot\bar{b}) + |\bar{a}\times\bar{b}| = 1 + |\bar{b}|^2$

$\sqrt{3}\,|\bar{b}|\cos\theta + |b|\sin\theta = 1 + |\bar{b}|^2$

$$\underbrace{\sqrt{3}\cos\theta + \sin\theta}_{\leq 2} = \underbrace{|b| + \frac{1}{|b|}}_{\geq 2}$$

$\therefore \quad \theta = \frac{\pi}{6} \qquad \therefore \quad \operatorname{cosec}^2\theta = 4$